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Crime Measures and Housing Prices: an Analysis Using Quantile Regression and Spatial Autocorrelation

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Abstract

Crime is a disamenity, so buyers should be willing to pay more for a house (all else equal) in a low crime area, suggesting that high crime rates depress housing prices. Conversely, it is plausible that criminals prefer wealthier areas because of the higher expected returns from their transgressions. This study examines the link between measures of crime and prices of residential housing. Our data begin in 2008 and end in 2020 for Seattle, Washington, including all reported felonies (756,304); 911 calls (1,528,303); all recorded residential real estate transactions (61,902 after filtering), as well as the corresponding property characteristics; demographic data and the associated changes. The inherent endogeneity between crime rates and housing prices forces us to find an instrument for crime rates. After rejecting several plausible choices, based on the Wu-Hausman test, the key variable in our empirical analysis is the number of 911 calls (contemporaneous and lagged) in a given beat. Our results present somewhat mixed evidence on the impact of crime on housing prices. Without adjustment for spatial autocorrelation, a 1 percentage point increase in crime rates (instrumented by the number of 911 calls) leads to approximately a 0.55% *decrease* in house prices. However adjusting for spatial autocorrelation changes this figure to a 0.80% *increase*. We further show that distance to a crime hotspot is significantly negatively related to housing prices, suggesting that criminals choose to operate in wealthier areas, which is consistent with our findings incorporating spatial autocorrelation.

Keywords Crime rates · Crime hotspots · Hedonic models · Quantile regression · Spatial autocorrelation · Wu-Hausman test

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Introduction

I never robbed any poor man yet, Nor any tradesman did I beset; I robbed both lords and the ladies bright, And brought their jewels to my heart's delight.
The Newry Highwayman (traditional Irish ballad)
Now we've seen a jump in violent crime. Criminologists can debate the causes of the new crime wave. But many people intuitively understand that places in which decay and disorder become the norm are places where crime tends to thrive. That's because crime is largely a function of environmental cues — of the palpable sense that nobody cares, nobody is in charge, and anything goes. We now live in a broken-windows world.
Brett Stephens, Our Broken Windows World, *NY Time*s, August 24, 2021

The social cost of crime has been extensively studied for centuries; its theoretical roots can be found in the work of Guerry[1] and Quetelet (1833), as well as in research in the early twentieth century by Arthur Pigou and others. An important later contribution is Becker (1968), which addresses the issue of the cost of crime and how resources should be allocated to enforcement. These costs, in particular indirect, intangible or long-term costs, are extremely difficult to measure, but are vital to the establishment of efficient policies for prevention, enforcement etc.[2] Within our context of real estate economics, the influence of (perceptions of) crime rates and changes in crime rates on residential housing prices (typically via a hedonic model using property and local characteristics) has been the subject of a myriad of studies across several disciplines: real estate, urban economics, criminology, public policy, social science etc. Overall the empirical research supports the notion that higher crime rates depress housing prices, but the results are relatively weak. For example, Ihlanfeldt and Mayock (2010b) documents that a 1% increase in the amount of violent crime leads to a 0.2 to 0.6% decrease in house prices, while property crime had no significant impact on housing prices.[3]

The absence of strong support for crime as a disamenity is in large part due to the formidable econometric obstacles that must be overcome. While these are discussed further in the next section, the major issues are: endogeneity, spatial autocorrelation, under-reporting of crime, the differential impact of various types of crime, the relative difficulty in obtaining sufficiently large data sets and the impact of demographic characteristics, such as population age or income levels.

It is reasonable to expect that consumers would view crime as a negative externality and thus a significant factor in their housing decisions: buyers should be willing to pay more for a residence in a low crime area, all else equal. Popular websites such as CrimeReports and SpotCrime contain information on crimes at the street address level so this information is readily available to potential buyers. Conversely,

[1] E.g., Guerry et al. (1833).
[2] Wickramasekera et al. (2015) provides a global survey of the cost of crime. Cohen (1990) uses jury awards to estimate the costs of different types of crimes.
[3] See Table 1 in Williamson and Ceccato (2015) for a summary.

it is plausible that criminals would opt to conduct their odious mischief in regions where the expected returns are higher (as the first epigraph suggests).

Based on these general issues and the related references reviewed in the following section, we have the following null hypotheses:

H_1: Alternative measures of crime do not influence residential property prices.
H_2: Demographic characteristics do not influence residential property prices.

Our empirical analysis is based on Seattle data from 2008 to 2020. We estimate hedonic models with and without adjustments for spatial autocorrelation. We believe our study can provide new insight into this important topic for a number of reasons. Firstly, our data cover a relatively long time period with a large number of observations. Our analysis of demographic data allows us to analyze the ramifications of levels and changes in demographic characteristics across each census tract. Secondly, our econometric techniques and unique data (in particular, quantile regression, our treatment of spatial autocorrelation and endogeneity and our use of 911 calls) hopefully can offer a new perspective.[4]

In our empirical analysis we find that the endogeneity of crime rates and housing prices is a significant issue. Several of our potential proxies for crime rates fail to pass the Wu-Hausman endogeneity test, which shows that the number of crimes (total, property and social) and changes in crimes (total and property) are all endogenous. We therefore use the number of 911 calls as an instrument in a two-stage least squares (2SLS) framework. The only other variables we found that appear to be exogeneous are: the number of violent crimes, the change in the number of crimes (violent and social) and the distance to a crime hotspot, discussed below.

In our hedonic models we estimate that crime rates (instrumented by the number of 911 calls) have an ambiguous influence on house prices. Specifically, a 1 percentage point increase in crime rates (instrumented by the number of 911 calls) leads to approximately a 0.55% *decrease* in house prices, but a 0.80% *increase* when spatial autocorrelation adjustments are added. When we restrict our 911 calls to only those with high priority, 68.1% of the total sample, the corresponding figures are 0.65% *decrease* and 0.21% *increase*, respectively.

We obtain more consistent results using a second variable that passes the Wu-Hausman test, the distance to a crime hotspot, where the clustering of crime is determined by the Getis-Ord statistic. This factor has a *negative* impact on housing prices, which suggests that criminals choose to operate in wealthier areas. Specifically, moving one mile farther from a crime hotspot in a given census tract leads to a 7.08% (6.28%) decrease in house prices in that tract without (with) adjustment for spatial autocorrelation. This result is in agreement with the above mentioned findings incorporating spatial autocorrelation.

[4] To the best of our knowledge, only three of the papers that analyze crime and housing prices incorporate spatial autocorrelation effects. These studies are discussed further in "Studies Incorporating Spatial Autocorrelation and Endogeneity" section.

Table 1 Number of reported crimes in Seattle

Panel A: Number of reported crimes per year

Year	Type of crime			Total
	Property	Violent	Social	
2008	42,505	9,429	2,440	54,374
2009	43,935	9,517	2,229	55,681
2010	41,441	9,406	2,442	53,289
2011	39,832	9,239	2,060	51,131
2012	39,875	9,162	1,781	50,818
2013	44,719	8,908	2,050	55,677
2014	48,637	9,425	2,222	60,284
2015	46,007	9,828	3,293	59,128
2016	47,283	10,328	4,016	61,627
2017	47,341	10,955	5,153	63,449
2018	48,357	12,060	5,441	65,858
2019	45,945	12,083	4,851	62,879
2020	47,748	11,069	3,292	62,109
N	583,625	131,409	41,270	756,304
Mean	44,894	10,108	3,175	58,177
Median	45,945	9,517	2,442	59,128

Panel B: Number of reported crimes per year by crime type

Property crime

Year	Arson	Burglary	Vandalism	Larceny	Robbery	Stolen property	Motor vehicle theft
2008	85	6,551	7,050	23,042	1,667	259	3,851
2009	148	6,641	6,375	25,355	1,795	225	3,396
2010	113	6,448	5,996	23,632	1,428	235	3,589
2011	82	6,772	5,723	22,141	1,420	248	3,446
2012	75	6,525	5,664	22,270	1,447	213	3,681
2013	87	7,308	5,718	25,501	1,602	250	4,253
2014	72	7,133	5,766	28,344	1,580	248	5,494
2015	92	7,564	5,977	26,616	1,546	340	3,872
2016	102	7,524	6,318	27,206	1,555	391	4,187
2017	150	7,802	6,643	27,003	1,539	495	3,709
2018	112	8,106	6,664	26,971	1,701	709	4,094
2019	112	7,715	5,833	26,107	1,557	626	3,995
2020	167	10,452	6,674	23,409	1,489	617	4,940

Violent crime

Year	Assault	Homicide	Kidnapping	Sex offenses	Sex offenses (Consensual)	Human trafficking
2008	9,009	30	55	318	17	
2009	9,102	21	54	323	17	
2010	9,037	17	45	293	14	

Table 1 (continued)

2011	8,845	25	48	304	17	
2012	8,739	23	57	330	13	
2013	8,537	19	46	291	15	
2014	9,016	23	55	319	12	
2015	9,394	27	40	358	9	
2016	9,780	19	58	461	10	
2017	10,311	27	77	521	18	1
2018	11,316	32	80	617	15	
2019	11,201	36	125	698	7	16
2020	10,335	53	128	534	5	14

Social crime

Year	Curfew	Prostitution	Trespassing	Weapon law violations
2008	103	302	1,280	755
2009	121	328	1,174	606
2010	125	572	1,164	581
2011	76	433	939	612
2012	55	250	956	520
2013	58	221	1,233	538
2014	42	313	1,371	496
2015	47	343	2,187	716
2016	62	457	2,826	671
2017	60	385	3,910	798
2018	64	122	4,408	847
2019	46	230	3,921	654
2020	22	29	2,629	612

Panel A reports the number of reported crimes by year and crime type from 2008 to 2020. Panel B partitions crimes into property (arson, burglary, vandalism, larceny, robbery, stolen property and motor vehicle theft), violent (assaulting, homicide, kidnapping, sex offenses and human trafficking) and social (curfew, prostitution, trespassing and weapon law violations) and reports the total number of crimes within each category by year

We also run the regressions using the other variables that pass the endogeneity test. The number of violent crimes has a significant negative effect in all of our models. This result is weaker when we use changes in violent crimes. The change in the number of social crimes has a weak positive influence. When we re-run the regressions using lagged values for the number of crimes (instrumented by lagged 911 calls) or those for violent crimes, we obtain qualitatively similar results.

Furthermore, in our hedonic models we find several demographic factors that significantly influence housing prices: (positively) levels and changes in White population; levels and changes in old population (% over 65); levels and changes in Asian population; population levels; (negatively) change in population; change in old population. Our quantile regression estimates generally demonstrate that the impact of crime is stronger for lower priced housing.

The remainder of this paper is organized as follows. "Literature Review" section presents a very brief review of the literature most relevant to our topic, focusing on the econometric issues. "Data and Methodology" section describes our data set and methodology. "Empirical Results" section summarizes our empirical results and "Conclusions" section presents our conclusions.

Literature Review

Econometric Issues

Early studies of the relation between crime rates and housing prices provide mixed results. Thaler (1978) finds crime increases housing prices, while Hellman and Naroff (1979) and Rizzo (1979) observe the opposite. While providing some valuable insight into this issue, these initial empirical works did not address the critical endogeneity problem: do high crime rates depress housing prices or do high property prices increase crime rates? Criminals may be attracted to high income areas because of more lucrative potential gains, raising crime rates in these regions.[5] Conversely, residents in wealthier areas have greater resources available to safeguard their assets, thus lowering crime rates.[6] A similar result would obtain if low-income residents tend to commit crimes in their own neighborhoods.[7] Exacerbating this endogeneity problem is the tendency for high crime areas to receive increased policing efforts, so called *hotspot policing*, which is a program implemented by most major US cities.[8] Furthermore, greater crime rates tend to spur a higher response from the community, an effort that is supported by the US Department of Justice through numerous crime prevention grant programs.

A second serious econometric issue is spatial autocorrelation.[9] For example, the errors from a hedonic pricing model for nearby residences should be positively correlated since the residences' unobservable characteristics are very likely similar. We analyze this subject in detail below.

A further difficulty with studies in this area is the under-reporting of crime, creating measurement error.[10] Skogan (1984) provides a global (though somewhat dated) survey, which, for example, documents robbery reporting at about 56%. Goudriaan et al. (2006) cite non-reporting figures as high as two-thirds of all crimes. Of course differences across country's definitions and reporting of crimes make these types of calculations problematic. MacDonald (2001) finds that under-reporting of property crimes varies with the economic cycle: reporting increases in good economic

[5] Case and Mayer (1996).

[6] This and related issues are discussed in Hipp (2011a).

[7] Curry and Spergel (1988) documents the link between crime rates and poverty.

[8] Braga et al. (2014).

[9] This concept is usually considered to stem from the work of Michael Dacey in the 1950s, building on central place theory. For more recent overviews see, i.a., Getis (2010).

[10] An early example is Myers (1980).

periods. MacDonald (2002) analyzes under-reporting from both sides: by the victims and by police. Goudriaan et al. (2006) further shows the relevance of neighborhood characteristics on reporting behavior. Under-reporting is clearly linked with perceptions of police legitimacy, which is another well researched topic.[11] Ceccato (2015) discusses the under-reporting of vandalism.

Another econometric issue is the precise definition of crime rate.[12] The empirical studies address various categories of crime. A number analyze total crime rates,[13] which thus equates murder with minor crimes such as graffiti. Others rely solely on property crimes,[14] burglaries etc. In general the results vary significantly by crime type, but not in a systematic manner. For example, Gibbons (2004) determines that vandalism, but not burglary crimes, negatively influences housing prices; Ihlanfeldt and Mayock (2010b) finds that only violent crime has a negative influence on housing values; Ihlanfeldt and Mayock (2010a) documents that only robbery and aggravated assault have a significant negative impact on property values; Braakmann (2012) observes that anti-social behavior and violent crime depress housing prices, but drug crimes do not; Ceccato and Wilhelmsson (2020) finds that vandalism rates have the greatest impact on housing prices. One can easily hypothesize that vandalism has a strong impact because of its visibility. In addition, the "broken windows" literature in criminology posits that enforcing petty crime (the reverse of current policy in Seattle and many other major US cities) can lead to reduction in major crimes.[15]

A further obstacle that empirical researchers face is obtaining a sample spanning a sufficiently long time period. Given the illiquidity of housing markets this fact makes investigation of housing price dynamics problematic. For example, Gibbons (2004) uses London data from April 1999 to March 2001; the analysis in Boggess et al. (2013) is based on Los Angeles data from 1993 to 1997. A long time series can lead to more precise estimation since seasonality in crime rates has been documented.[16]

Finally, confounding the relation between housing prices and crime rates are demographic characteristics such as education, income, ethnic composition,[17] rent versus own, etc. The literature has shown that crime rates are related to measures of "social capital," the social bonds within a community, which generate higher levels of personal engagement that can deter crime.[18] This topic is especially relevant

[11] Tankebe (2013).

[12] Kelly (2000) explores the differing impacts of inequality and poverty on violent and property crimes.

[13] E.g., Hellman and Naroff (1979); Ceccato and Wilhelmsson (2020).

[14] Gibbons (2004).

[15] The concept of broken windows policing is particularly relevant in Seattle because of recent moves to curtail prosecution of most misdemeanors. For example, from a 2019 posting on King5.com: *System Failure: Report finds Seattle prosecutors don't pursue 46% of misdemeanors.* See also, from April 22, 2022: https://kuow.org/stories/why-is-seattle-dropping-two-thousand-misdemeanor-cases.

[16] I.a., Andresen and Malleson (2013).

[17] Sampson and Wilson (2020).

[18] This topic falls under the general rubric of collective resources. From Skogan (1989; p.438): … *the same factors that that generate crime may also erode the social basis for collective action.*

to our study since in 2015, Seattle initiated its Micro-Community Policing Plans (MCPP), discussed in "Crime Data Description" section. The importance of these societal factors is outlined in the seminal work on social disorganization: Shaw and McKay (1942); see also Glaeser and Sacerdote (1999). These dynamics are also influenced by "endogenous gentrification" as analyzed in Guerrieri et al. (2013) and the related work on urban decline in Guerrieri et al. (2012).

Key Urban Economics Literature

An influential study in this area is Thaler (1978), which finds a positive relation between crime rates and housing prices.[19] Later research typically finds that crime tends to depress housing prices. However, as noted above, these early studies do not properly address the endogeneity issue. This topic is detailed in Ihlanfeldt and Mayock (2010b), which examines 18 studies from 1960 and 2006 and concludes that only two have adequately addressed this econometric issue.

More recent studies, after correcting for endogeneity (usually via an instrumental variable approach), generally find a negative, but usually weak relation between crime rates and housing prices.[20] Ceccato and Wilhelmsson (2018) reviews 31 studies and concludes that more than two-thirds document a negative relation between crime and property values. This reference also explores the relation between accessibility (e.g., through public transportation) and crime rates. For example, good accessibility may increase housing values while simultaneously increasing crime rates; the very early literature documents a negative relation between crime rates and the distance from the central business district.

An analysis of the influence of crime that does not focus on prices is Boggess et al. (2013). It argues that the hedonic model approach understates the impact of crime since it does not accurately assess the effect of crime on housing turnover. It finds that higher crime rates in high vacancy neighborhoods lead to less frequent transactions and that violent crime has significantly greater impact than total crime.

The connection between housing and macroeconomic characteristics and crime rates has also been extensively examined. For example Roncek (1981) and later studies find a negative relation between home ownership and crime rates. Cullen and Levitt (1999) verifies that increased crime rates lead to decreased population (outmigration) rates. Hipp (2011b) and others have investigated the interaction between crime rates and ethnic composition. Pyle (1998) and many others have analyzed the criminogenic nature of unemployment. Roncek and Maier (1991) establishes a positive connection between the number of liquor establishments and crime rates. Finally, "fear of crime" has been shown to depress housing prices.[21]

[19] Other examples include Hellman and Naroff (1979) and Lynch and Rasmussen (2001).

[20] For example, Schwartz et al. (2003); Ihlanfeldt and Mayock (2010a); Ceccato and Wilhelmsson (2011); Braakmann (2012).

[21] An interesting early reference is Wilson and Kelling (1982). A comprehensive review is found in Hale (1996). Larsen et al. (2003); Linden and Rockoff (2008); Pope (2008) analyze the significant negative influence of sex offender registration on nearby housing prices.

Studies Incorporating Spatial Autocorrelation and Endogeneity

As mentioned earlier, a potential problem with hedonic models for housing prices is spatial autocorrelation: because of similarities in omitted variables or unobservable characteristics, it is likely that model errors for nearby residences are positively correlated. A comprehensive treatment is given in Pace and LeSage (2009). Zietz et al. (2008) accounts for this effect by using contiguity matrices, which define neighbors as those regions having a common side. In the weighting matrix, contiguous observations are coded as 1, 0 otherwise. Zhang (2016) handles the problem by including a spatial weight matrix, where the weights are based on the normalized Euclidian distance between a house and its neighbors. A regression parameter is then estimated in the hedonic model regression. This methodology is described in the following section.

The spatial autocorrelation concept is related to the large body of research that discusses the clustering of crime into *hotspots*, regions that generate a higher proportion of crimes than similar areas.[22] This approach is plausible since it is unlikely that crime rates are uniformly distributed over the sample area, such as a census tract.[23] Previous research has shown that the identification of hotspots is sensitive to the definition of a region. For example, correlations appear to increase as the level of aggregation grows.[24]

To the best of our knowledge, there are 3 papers dealing with crime rates and housing prices that adopt an econometric approach similar to ours.[25] The first is Wilhelmson and Ceccato (2015), which uses spatial autocorrelation and quantile regression to analyze the impact of crime on apartment prices in a nonurban area, Jönköping, in southern Sweden. This paper also formally addresses the endogeneity issue. It finds that burglaries had no impact on prices in 2005, but a significant negative impact in 2011, in addition, distance to the city center had a significant negative impact on prices.

McIlhatton et al. (2016) analyzes different types of crime and housing prices in Belfast over the period 2012 to 2014. This study documents positive spatial autocorrelation of crime rates and uses instruments to account for endogeneity. It shows that crime rates (in particular, burglary) are positively related to housing prices while other types of crime (personal violence, drugs etc.) are associated with lower priced housing.

[22] The initial references are usually credited to Brantingham and Brantingham (1982) and Sherman, Gartin and Buerger (1989). An overview of crime spatial analysis is presented in Anselin et al. (2000). Weisburd (2015) provides an overview of the empirical data on hotspots and crime concentration.

[23] Weisburd, Bushway, Lum and Yang (2004) reports that half of the reported crimes in its Seattle data come from less than 5% of the total locations. A significant body of research develops the rationale for the existence of hotspots, in particular *contagious diffusion*, where criminal activity in one area affects adjacent areas; See, i.a., Roncek and Bell (1981).

[24] See Gehlke and Biehl (1934) for a very early analysis.

[25] A study that incorporates spatial autocorrelation in an analysis of the Hong Kong apartment market is Hui, Chau, Pun and Law (2007).

More recently, de La Paz et al. (2022) studies crime and housing prices in LA County in 2016. The focus of this research is on the potentially nonlinear effects of crimes of varying types. It documents clearly the existence of crime hotspots across all types of crime studied. In this sample, transaction prices decrease significantly for aggravated assaults occurring within 2,000 m; 1,000 m for burglary; 500 m for drug crimes. Furthermore, crimes (especially aggravated assaults and burglary) increase in proximity to higher-priced houses, indicating that criminals choose to operate in wealthy areas. However, when an instrumental variable approach is adopted, the negative impact of crimes on house prices becomes insignificant for all types of crimes, except for narcotics and non-aggravated assaults.

These three studies are all important contributions to our understanding of the subtle links between crime rates and the pricing of residential housing. We hope that our analysis can contribute to this literature since it differs from these studies in several important ways. First, our sample covers a longer time period (2008 – 2020) with a larger number of observations. This is a significant difference because, as these papers and the related research show, the empirical impact of crime presents very significant econometric difficulties. In addition, our longer time series allows us to exploit the panel structure of our data by including month and year fixed effects, as well as changes in the underlying demographics.[26] Secondly, building on the statistical approaches used in these papers, we believe that we have created a methodology that provides a sound path to resolving the econometric problems: quantile regression to address the changing nature of the hedonic models across residences of varying quality; spatial autocorrelation adjustments to correct for the clustering of crime rates and the positive autocorrelation of adjacent hedonic model residuals; the Wu-Hausman test to formally address endogeneity issues; a precise definition of crime hotspots and coldspots using the Getis-Ord statistic; and, perhaps most importantly, the use of 911 calls as a strong instrument for crime rates.

Finally, no clear consensus emerges from the extant literature, and all of these studies (including ours) are subject to the criticism that the results are dependent on the geographic and anthropogenic characteristics of the specific areas being studied, so examining a different and interesting city is valuable.

Data and Methodology

Data

The data analyzed in this study come from several sources. We obtain data on property transactions from the King County Assessor's website, which provides transaction data for residential and commercial properties, as well as for lands, easements, etc. The property transactions data available for downloads date back to the 1950s; we use the data for single-family houses in the City of Seattle from 2008 through

[26] We also performed our regression analyses using zip code fixed effects and obtained very similar results (as reported in Appendix 2 Table 11) to those reported here.

2020 to match the availability of our crime data. We use the property transactions data to build a dataset for our hedonic variables: sale prices, square footage (of living area, lot area, etc.), number of bedrooms and full-size bathrooms, year of construction and indicators for natural views (Lake Washington, Mt Rainier, Puget Sound etc.).

We apply a number of selection filters to remove: (i) non-arm's length transactions, (ii) transactions where multiple residential properties are sold, (iii) transactions with sale prices below $10,000 and (iv) other filtering criteria commonly used in the real estate literature. Moreover, we deleted observations that were made in thin cells, defined as zip codes with relatively few home sales. For each year, we calculated the total number of transactions per zip code, and excluded zip codes with a transaction volume lower than the 10th percentile. We further eliminated observations from zip codes that did not lie entirely in Seattle. In total, we have 61,902 single-family residential property sales in the City of Seattle over the January 1, 2008 – December 31, 2020 period. This filtering also reduces the number of census tracts analyzed in our study to 105.

Demographic data are collected from the Washington State Office of Financial Management. The data are available on an annual basis and provide, i.a., population, age, gender and race at the census tract level. Finally, we collect data on all reported crimes in the City of Seattle from the Seattle Police Department (SPD) website. The data are available for 2008 through 2020 and consist of dates, locations (with longitudes and latitudes[27]) and types of crime (violent, property or social) reported to the SPD during the same period. We thus have a total of 756,304 reported crimes in Seattle in our sample.

Crime Data Description

Seattle provides an ideal location for the analysis of the influences of crime.[28] The city has been the subject of a substantial number of related studies. For example, Weisburd et al. (2004) examines the distribution of crime in Seattle from 1989 to 2002 showing that relatively small areas are responsible for the majority of shifts in crime rates. Matthews et al. (2010) documents significant levels of spatial clustering of property crime in Seattle over the 1998 to 2000 period. Kreager et al. (2011), using Seattle data from 1980 to 2000, shows that gentrification initially leads to small increases in crime, but subsequently is correlated with decreases in crime. Velez et al. (2012), using Seattle data from 1980 to 2007, provides evidence that increased mortgage volume leads to subsequent decreases in crime.

In January 2015 Seattle introduced its Micro-Community Policing Plans (MCPP) designed to enhance police efficiency through a collaboration between police and

[27] When we aggregate crime data at the census tract level, we use reverse geocoding to convert longitudes and latitudes of crime locations to street addresses. Reverse geocoding is done at Texas A&M University Geoservices (http://geoservices.tamu.edu).

[28] "*More than a quarter-million residents of greater Seattle have considered leaving the Emerald City due to rising crime — the highest percentage of any large metro area in the country, according to a new national survey.*" New York Post, October 6, 2023.

community. Its goal was to adapt police procedures to the individual region's characteristics. Perceptions of crime and related topics are annually surveyed in the Seattle Public Safety Surveys. Furthermore, policing was perhaps the major issue in the November 2021 Seattle mayoral elections.[29]

We use data on each reported felony from 2008 to 2020; we do not include white collar crimes (such as tax evasion), drug offences or traffic violations. We analyze the detailed demographic data for each census tract. Residential real estate transactions are available over our entire sample period. The abundance of crime provides us with a rich data set.[30]In addition, a large literature has emphasized the higher crime rates in urban locations.[31]

Table 1 presents a summary of our 756,304 crime observations over 2008 to 2020. Panel A partitions crimes into property (arson, breaking and entering, property damage, vandalism, larceny/theft, motor vehicle theft, robbery and stolen property offences), violent (assault, homicide, human trafficking, kidnapping and sex offences) and social (curfew, loitering and vagrancy offences, prostitution offences, trespass and weapon violations). Overall there is a mildly increasing trend; the 3 categories showing annual growth rates of 0.9%, 1.2% and 2.3%, respectively. As expected, the different types of crime are positively correlated: the correlation between property and violent crime is 0.616; between property and social crimes is 0.648; between violent and social crimes is 0.919. Panel B contains details on property, violent and social crimes per year. For violent crime, there are positive growth rates for all 4 of the major categories. For property crime, positive growth rates are evident for arson (5.79%), burglary (3.97%), stolen property (7.50%) and motor vehicle theft (2.1%). Trespassing is the only social crime with a positive growth rate (6.18%).

Figure 1 shows a choropleth map of crime intensity in Seattle in 2010, 2015 and 2020. These diagrams clearly show the concentration of crime around the Westlake Center and, in particular, in 2020, its rapid expansion into the University of Washington area. Figure 2 shows the corresponding dynamics of housing prices over our sample period. The growth in the sale of high-end residences is apparent, as are the higher valuations for waterfront properties.

911 Calls

We construct an instrumental variable, *911 Calls*, based on the number of 911 calls made in a given year. We obtain these data from the Seattle Open Data Portal on the City of Seattle's webpage,[32] which records all calls made to the Seattle Police Department. The data are available from 2009 through 2022. In the dataset, each call is time stamped and categorized by call type (911, non-911, complaint, etc.), and the

[29] Policing a key issue as Seattle chooses new mayor, AP, Nov. 2, 2021.

[30] In a 2019 WalletHub study Seattle's crime rate ranked 150 highest out of 180 surveyed US cities. A 2020 FBI survey ranked Seattle first out of the 30 largest metropolitan areas in overall property crime.

[31] E.g., Glaeser and Sacerdote (1999).

[32] https://data.seattle.gov/.

call origination location is provided. Per the Seattle Police Department's intention to safeguard the privacy of each caller, latitude and longitude specifics of call origins are not reported. Instead, calls' originated locations are recorded at the "beat" level; beats are the patrol areas that individual officers have responsibility for. There are 54 beat areas within the City of Seattle. We have 1,528,303 911 calls made to the police department from 2009 to 2020.[33] Finally, we construct the *911 Calls* variable as:

$$911\ Calls_t = \log\left(1 + number\ of\ 911\ calls\ per\ beat\ in\ year\ t\right).$$

Property and Demographic Data

Table 2 provides summary statistics for our key explanatory variables based on our sample of 61,902 sales from January, 2008 to December, 2020. The mean (median) sale price is $706,110 ($617,500). The property-level statistics show a mean (median) living area of 1,891 (1,720) sq. ft.; lot area of 4,660 (4,650) sq. ft.; number of bedrooms 3.2 (3.0). Residences with a view (principally of the Olympic or Cascade Mountains) represent 12.1% of the total. The clustering of crime hotspots is apparent from measuring the distance from the nearest to the 5*th* nearest: 3.8 to 4.4 (in straight distance miles); 4.8 to 5.8 (in travel distance miles); 12.3 to 15.5 (in travel minutes).

Table 3 depicts the time series of our crime and demographic data. It shows that crime per capita has been relatively unchanged over our 2008 to 2020 sample period: the mean rate has dropped very slightly (−0.2%) while the median rate has increased by a modest 0.34%. This lack of a significant trend suggests that the change in crime rate variable, which is used in many studies in order to mitigate endogeneity, will be unlikely to play a meaningful role in our analysis. In addition, the most recent report by Seattle Police Department (2023) shows that there is no notable seasonal trend in city-wide crimes. The average number of 911 calls, on the other hand, has had a steady increase over time, increasing from 1,153 calls per beat in 2009 to the peak of 2,940 calls in 2014. Population per tract has increased by 21%. The average percentage of White, Asian, Black and Old (defined as over 65) has changed by −6.16, 3.89, 0.80 and 1.67%, respectively, reflecting some of the pivotal shifts in Seattle's population mix.

Methodology

Hedonic Models

Our empirical analysis of prices starts from a familiar point: a spatial hedonic model weighting the physical characteristics of the property and the specifics of its

[33] Since 911 calls data are available from 2009, our main analysis covers the period from 2009 to 2020. Crime and demographic data are available from the pre-2009 period; we use crime and demographic data from 2008 to construct their lagged variables and associated annual change variables.

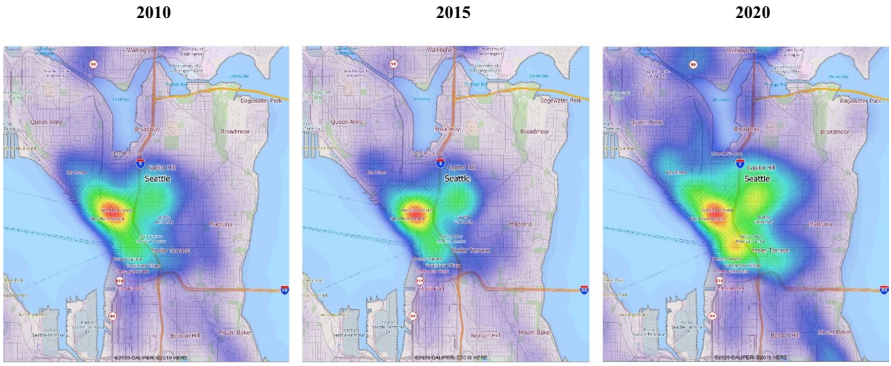


Fig. 1 Crime hotspots in Seattle (2010, 2015 and 2020). This figure shows choropleth maps of crime intensity in Seattle. The displayed density maps provide a visual representation of total crime counts reported to the Seattle Police Department for the years 2010, 2015 and 2020. To create these maps, we use GIS software to plot the location (based on XY coordinates) of each reported crime onto the city's layout. The plotted points are then transformed into a regular grid cell system. Within each grid, each cell is assigned a value based on the density of nearby crime points. The resulting density at any given location is visually represented by a color gradient: blue indicates low crime density, progressing through green and yellow, with red signifying high crime concentration areas

neighborhood.[34] This technique has applications far beyond real estate, but in our context the approach probably originates with Kain and Quigley (1970). Since then there have been numerous important contributions, which are detailed in several surveys, including: Malpezzi (2003); Sirmans et al. (2005); Herath and Maier (2010).[35]

The results of the estimations for housing prices in the numerous hedonic models in the literature vary considerably, as pointed out in Sirmans et al. (2005). Our basic approach is to add crime and demographic variables to the hedonic models to test for their relative significance, i.e., the implicit (hedonic) price of the attribute. In this manner both the tangible and intangible costs of crime can be estimated.[36]

A relatively recent refinement of the initial hedonic model approach incorporates quantile regression. Quantile regressions allow the *conditional* distribution of the dependent variable to be determined as a function of the independent variables. It is particularly useful when the explanatory variables have a different impact in the tails of the dependent variable's distribution than in the center of the distribution. Unlike running separate regressions for each segment (e.g., stratified by price), quantile regressions use the entire sample.

[34] Goodman (1998) provides a summary of the early work of Andrew Court on hedonic pricing models.

[35] Herath and Maier (2010) provide some further motivation for our study (from the abstract): *In general, implicit value of environmental amenities in the neighbourhood and air pollution are relatively overresearched. The effect of social factors, i.e. racial segregation and crimes on real estate value is underresearched.*

[36] A more prurient example of hedonic pricing using King County data is Brooks et al. (2020).

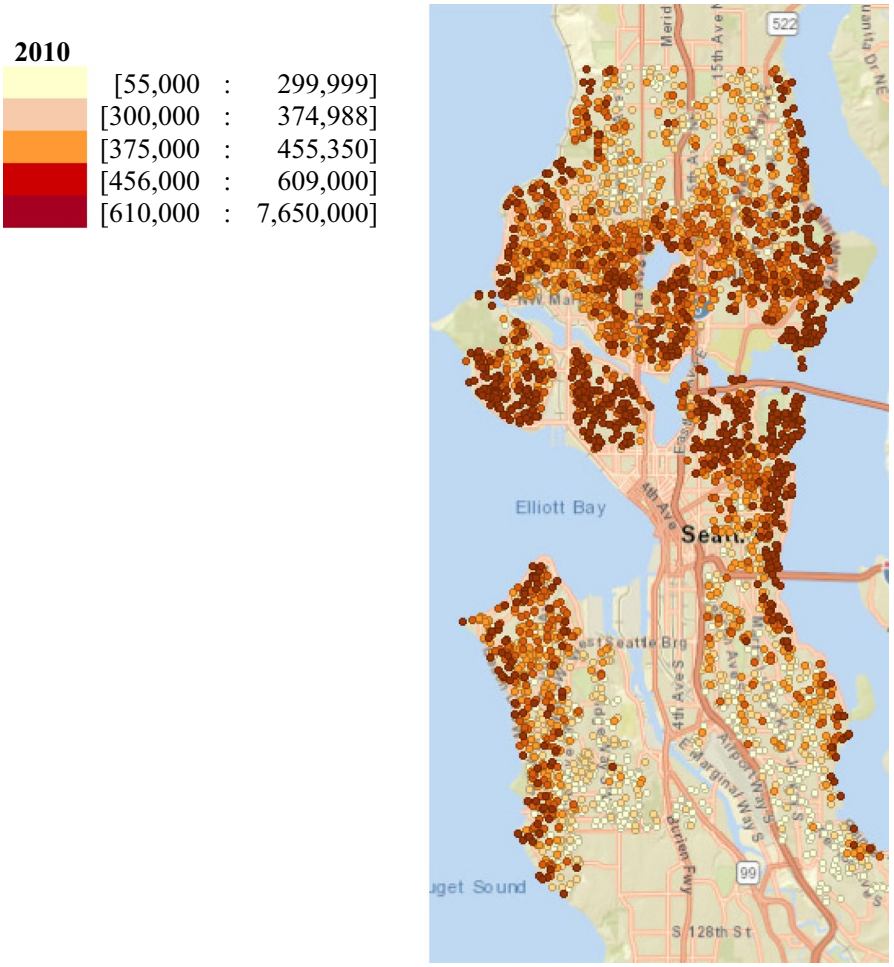


Fig. 2 Heat map of home sale prices in Seattle (2010, 2015 and 2020). This figure reports locations of single-family home sales ($) in Seattle in 2010, 2015 and 2020. Each dot represents the location of a house sale. In each year, home sales are divided into quintiles based on the sale price

It has been shown that the market for residences is segmented: the market for lower priced residences has very different dynamics than that for more expensive housing. A good example of its application is Zietz et al. (2008), which uses quantile regression to show that different characteristics of a property (such as living area, age and number of bathrooms) are not priced identically across the entire distribution of house prices.[37] We also adopt this approach.

[37] Koenker and Hallock (2001) provides an overview of the technique. See Deng et al. (2014) for an application to Singapore commercial real estate.

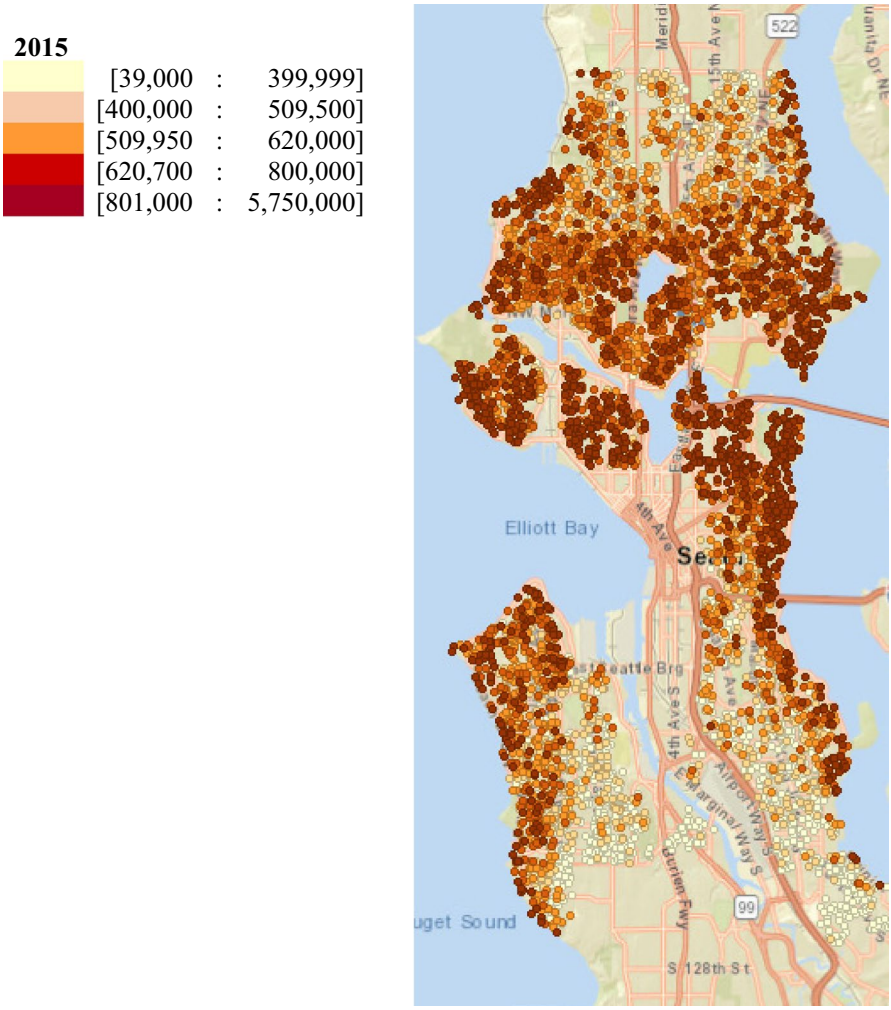


Fig. 2 (continued)

Spatial Autocorrelation

A more recent extension of the hedonic model methodology incorporates spatial autocorrelation, which attempts to account for the (assumedly positive) correlation between residuals from the hedonic model for neighboring residences. Zhang (2016) uses an adjustment for spatial autocorrelation to estimate the influence of flood plains on residential housing prices. Our approach is quite similar to hers.[38]

[38] A different approach is taken in Waltl (2019), which uses penalized quantile regression together with hedonic imputation (modeling unobserved prices) to analyze aggregate price movements across different market sectors.

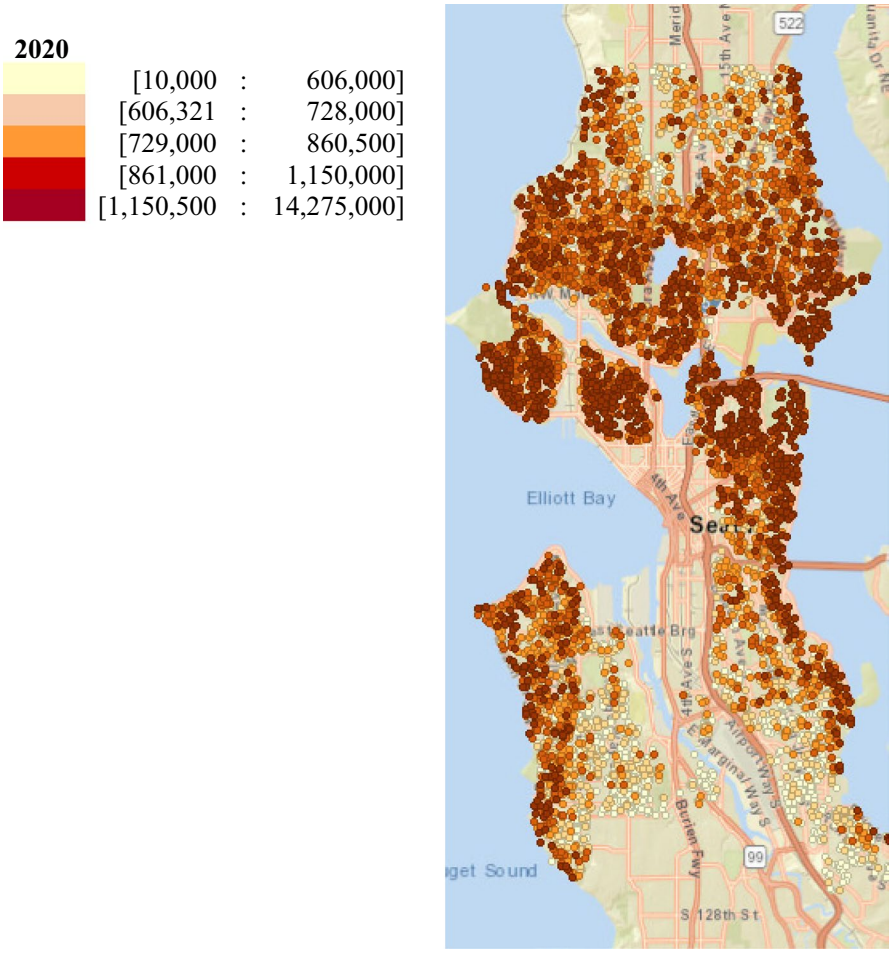


Fig. 2 (continued)

Our approach to capture the relation between neighboring observations follows that described in Getis (2010: p. 256):

> If a matrix Y represents all of the ($n^2 - n$) associations between all realizations of the Y variable in region R and W represents all of the ($w^2 - w$) associations of the spatial units to each other in region R, irrespective of Y, then the degree to which the two matrices are positively (negatively) correlated is the degree of positive (negative) spatial autocorrelation.

More specifically, $[Wy]_{it}$ is a spatial weight matrix, defined as $\left[Wy\right]_{it} = \sum_{j=1}^{n} w_{ij} y_j$ where y_j is the sales price for a "nearby recently sold" house j; w_{ij} is the weight (or spatial lag) assigned to y_j based on the inverse distance between house i and house j. Here the usual candidates for inverse distance calculation are: (i) $\frac{1}{d}$; (ii) $\frac{1}{d^2}$; (iii) $\frac{1}{d^5}$.

Table 2 Summary statistics

	N	Mean	Median	Std Dev	Min	Max
Property-level characteristics						
Sale price	61,902	706,110	617,500	457,343	10,000	14,275,000
WY	61,902	657,449	592,119	324,996	74,027	6,257,625
Living area	61,902	1.891	1.720	0.792	0.350	4.590
Lot area	61,902	4.660	4.650	2.636	0.375	14.466
Bedrooms	61,902	3.192	3.000	0.921	0.000	11.000
Bathrooms	61,902	1.485	1.000	0.689	0.000	7.000
One story	61,902	0.398	0.000	0.490	0.000	1.000
Age	61,902	55.571	65.000	38.399	0.000	114.000
Renovated	61,902	0.048	0.000	0.214	0.000	1.000
FirePlace	61,902	0.712	1.000	0.453	0.000	1.000
Environmental—natural views						
Lake/Mountain view	61,902	0.121	0.000	0.327	0.000	1.000
Distance to crime hotspots						
Straight distance (miles)						
Nearest	61,902	3.784	3.967	1.609	0.031	8.140
2nd nearest	61,902	4.015	4.199	1.595	0.299	8.165
3rd nearest	61,902	4.140	4.332	1.592	0.445	8.334
4th nearest	61,902	4.294	4.478	1.609	0.592	8.645
5th nearest	61,902	4.428	4.604	1.607	0.614	8.728
Travel distance (miles)						
Nearest	61,902	4.795	5.088	1.962	0.046	10.902
2nd nearest	61,902	5.195	5.506	1.979	0.476	11.156
3rd nearest	61,902	5.428	5.745	2.021	0.638	11.348
4th nearest	61,902	5.595	5.891	2.014	0.755	11.374
5th nearest	61,902	5.802	6.114	2.011	0.825	11.396
Travel time (minutes)						
Nearest	61,902	12.335	12.864	3.992	0.118	27.848
2nd nearest	61,902	13.544	14.093	3.646	3.030	27.877
3rd nearest	61,902	14.076	14.627	3.614	3.412	28.819
4th nearest	61,902	14.835	15.399	3.546	4.221	29.700
5th nearest	61,902	15.488	16.001	3.556	4.553	29.980
Sociodemographic characteristics (census tract-year observations)						
	N	Mean	Median	Std Dev	Min	Max
Population	1,235	5,138	4,900	1,683	1,915	11,930
% Male	1,235	0.492	0.491	0.018	0.439	0.567
% Female	1,235	0.508	0.509	0.018	0.433	0.561
% White	1,235	0.709	0.792	0.194	0.128	0.922
% Asian	1,235	0.139	0.095	0.113	0.032	0.639
% Black	1,235	0.083	0.033	0.097	0.005	0.426
% Old population	1,235	0.116	0.113	0.041	0.040	0.256

The sample consists of 61,902 single-family home sales in Seattle from 2008 to 2020. Definitions of the variables are provided in Appendix 1 Table 10

The second choice seems the most frequent choice in the related literature and the one that we employ in our analysis.

To calculate Wy for home i's sale (shown as a red dot • in Fig. 3), we first identify sales of its neighbor homes ($j=1,2,\ldots$) that occurred within 2,000 feet, within the past 24 months. The dark dots within the red circle are the sales of neighbor homes. For each neighbor home sale j, we measure the distance from home sale i and use the sale price y_j for neighbor home j to calculate $w_{ij}y_j$. Finally, we calculate the spatial weight matrix for the sale of house i, $[Wy]_{it}$, using the above equation.

Hotspot and Coldspot Characterization

We define a crime hotspot as a census tract with high cluster cores (densities) of crimes.[39] As noted above, there is a substantial literature that documents the existence of crime hotspots, specifically in Seattle. For each year, we aggregate the number of reported crimes at the census tract level, and compute the Getis and Ord (1992) statistics, G_i^* for each tract i, defined as:

$$G_i^* = \frac{\sum_j w_{ij}x_j}{\sum_j x_j}$$

The numerator is the weighted average number of crimes reported in the neighboring locations (census tracts $j=1,\ldots,105$). The denominator is the sum of all crime counts across all the census tracts. We categorize a census tract with a G_i^* larger than the mean as a "high-high cluster," or "hotspot." Similarly, a census tract with a G_i^* lower than the mean is categorized as a "low-low cluster," or "coldspot." For each year we select census tracts defined as hotspots based on G_i^*, and sort them from the highest to lowest within the High-High cluster. A census tract with the highest G_i^* statistics (with a p-value <0.05 for significance) is defined as a hotspot, the tract with the highest crime density. We also define a census tract with the second, third, fourth and fifth highest and significant G_i^* statistics as the hotspot with the second, third, fourth and fifth highest crime density.

Figure 4 depicts census tracts with high crime density (red), those with low crime density (blue) or no significant density (gray) in 2010, 2015 and 2020. Those categorized as red and blue have G_i^* with p-value <0.05.

Endogeneity

A major confounding issue in this analysis is the likely endogeneity of crime rates and housing prices, as noted earlier. This problem was initially handled in the context of hedonic models by Burgess and Harmon (1991). To address this issue we use the Wu-Hausman test, which assumes the null hypothesis of no endogeneity.[40] It tests for the correlation between the regression residuals and a given explanatory

[39] Ceccato and Wilhelmsson (2020) shows that housing prices are more sensitive to distance to crime hotspots than to overall crime rates.

[40] Wu (1973) and Hausman (1978).

Table 3 Average number of crimes and sociodemographic variables per census tract by year

Year	# Crimes (Per Tract)		Crime per capita		# 911 calls (Per Beat)		Population per tract			
	Mean	Median	Mean	Median	Mean	Median	Mean	Median		
2008	386	312	0.088	0.070	-	-	4,603	4,425		
2009	395	341	0.088	0.072	1,153	1,159	4,642	4,493		
2010	384	315	0.085	0.072	1,947	1,929	4,667	4,540		
2011	367	305	0.080	0.067	2,148	2,158	4,703	4,559		
2012	357	292	0.078	0.063	2,460	2,511	4,722	4,590		
2013	398	317	0.085	0.067	2,830	2,893	4,819	4,640		
2014	428	352	0.090	0.069	2,940	2,927	4,882	4,702		
2015	411	335	0.083	0.068	2,887	3,026	4,992	4,760		
2016	422	343	0.083	0.070	2,818	2,827	5,137	4,858		
2017	429	331	0.082	0.065	2,877	2,800	5,309	5,008		
2018	452	335	0.085	0.065	2,787	2,617	5,421	5,142		
2019	417	317	0.077	0.060	2,749	2,502	5,474	5,142		
2020	475	395	0.086	0.073	2,316	2,161	5,565	5,234		
Year	% Male population		% White population		% Asian population		% Black population		% Old population	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
2008	49.51%	49.27%	72.43%	81.38%	13.45%	8.87%	8.13%	3.31%	10.78%	10.33%
2009	49.56%	49.27%	72.26%	80.93%	13.46%	9.12%	8.14%	3.41%	10.74%	10.18%
2010	49.51%	49.05%	72.31%	80.69%	13.48%	9.13%	7.95%	3.47%	10.74%	10.38%
2011	49.45%	49.06%	71.94%	80.45%	13.67%	9.16%	7.96%	3.33%	10.98%	10.57%
2012	49.41%	49.06%	71.48%	80.05%	13.69%	9.25%	8.22%	3.63%	11.64%	11.08%
2013	49.48%	49.11%	71.32%	79.72%	13.82%	9.28%	8.08%	3.44%	11.86%	11.38%
2014	49.50%	49.09%	70.54%	79.10%	14.29%	9.61%	8.20%	3.79%	12.03%	11.60%
2015	49.45%	49.16%	70.18%	78.51%	14.43%	9.90%	8.32%	3.87%	12.35%	11.95%
2016	49.40%	49.10%	69.26%	77.77%	14.94%	10.44%	8.59%	4.02%	12.04%	11.58%
2017	49.45%	49.11%	68.66%	77.14%	15.45%	10.95%	8.61%	3.99%	11.96%	11.54%

Table 3 (continued)

2018	49.59%	49.14%	68.17%	76.38%	15.86%	11.58%	8.62%	4.05%	12.00%	11.58%
2019	49.52%	49.14%	67.34%	75.45%	16.63%	12.28%	8.64%	4.16%	12.26%	11.76%
2020	49.50%	49.12%	66.26%	74.41%	17.34%	12.91%	8.93%	4.26%	12.44%	11.98%

We report average values for the number of crimes, crime per capita, number of 911 calls and sociodemographic variables measured at the census tract level. *# Crimes* is the number of total crimes (violent, property and social) reported in a census tract in a given year. *Crime per capita* is defined as *# Crimes per tract* divided by population in the same census tract. *# 911 calls* is the number of 911 calls made per police beat in a given year. *Population per tract* is the total population in a census tract. *% Male/White/Asian/Black population* is the percentage population of the respective gender/racial group divided by population in the same census tract. *% Old population* is the percentage population of people at age 65 or older divided by population in the same census tract

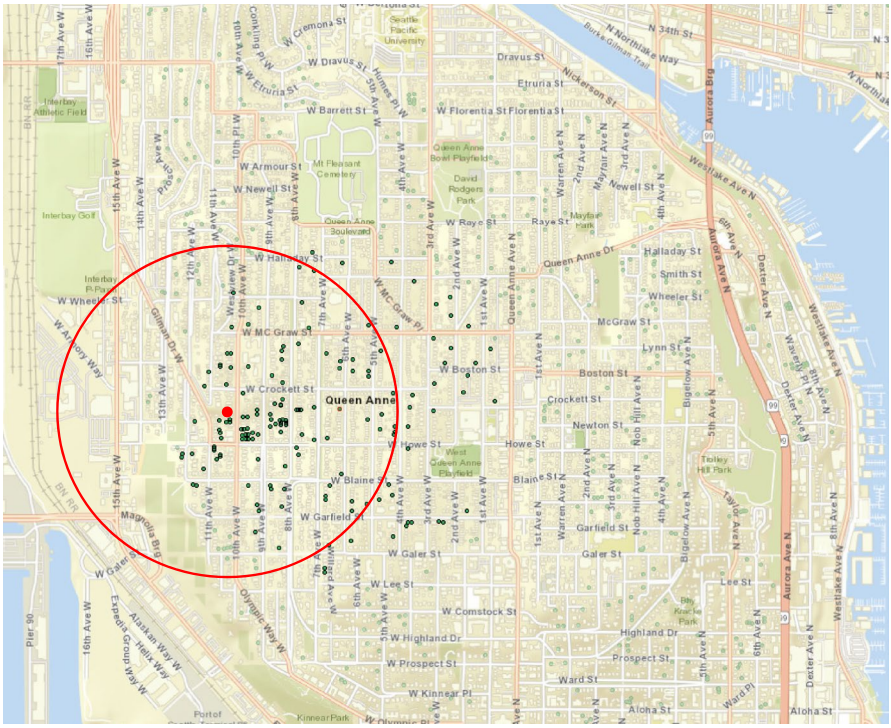


Fig. 3 Connectivity map for spatial autocorrelation weight. This figure shows a connectivity map illustrating the selection of nearby house sales used to compute the spatial weight matrix, as defined in "Hotspot and Coldspot Characterization" section. For each house sale in our sample, we identify sales of its nearby houses that occurred within the past 24 months. The circle is centered on a house sale and each dot within the circle represent a house sale made within 2,000 feet

variable. A sufficiently low p-value means that we reject the hypothesis of no endogeneity. In the case of rejection, we are compelled to use an instrumental variable approach, which the literature suggests is a non-trivial pursuit.[41]

As possible instruments for crime rates we use 911 calls, changes in crime rates and distance to a crime hotspot.[42] We then use the Montiel-Pflueger test to eliminate weak instruments.[43] More specifically, our (Durbin) Wu-Hausman test consists of two stages. In the first stage, the potential endogenous variable, *Number of Crimes* is regressed on: (i) a potential instrumental variable (e.g., *911 Calls*) and (ii) other right-hand side variables in the regression as estimated below in Eq. 1. The residual values from the first stage regression are then used as an additional regressor in the original OLS regression.[44] If the coefficient on the added regressor (i.e., residuals from the 1st-stage regression) is significant, *Number of Crimes* is determined

[41] E.g., Spelman (2013).
[42] Ihlanfeldt and Mayock (2010a) uses commercial land use as an instrument for crime rates.
[43] Olea and Pflueger (2013).
[44] This is called an Augmented (Durbin) Wu-Hausman test.

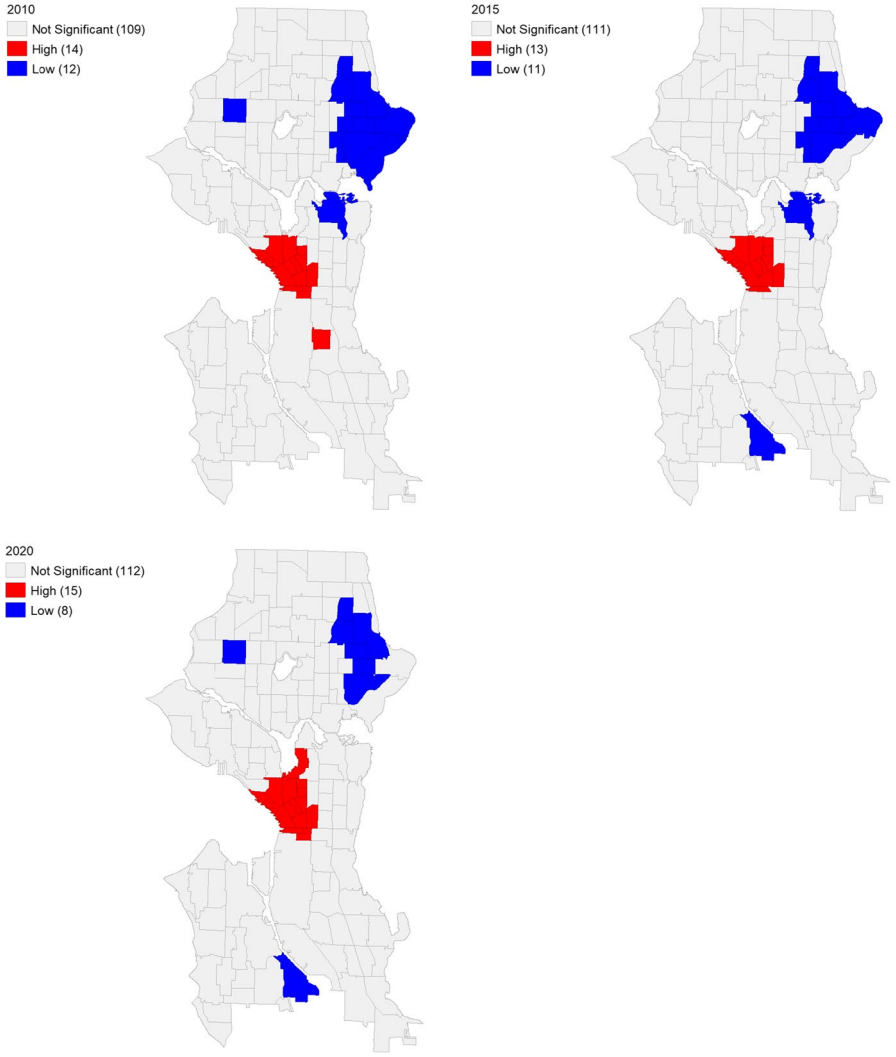


Fig. 4 Crime hot and coldspots in Seattle (2010, 2015 and 2020). This figure shows cluster maps for census tracts in Seattle with high-high cluster cores (hotspots) and those with low-low cluster cores (coldspots). For each census tract in each year, we aggregate the number of reported crimes at the tract level. We compute the Getis-Ord statistic (the G^* statistic) for each census tract based on its crime intensity, proximity to and crime intensity in its nearby census tracts. We define a census tract with a G^* statistic larger (p-value < 0.05) than the mean in a given year as a "hotspot (in red)". We define a census tract with a G* statistic significantly lower (p-value < 0.05) than the mean in a given year as a "coldspot (in blue)"

to be endogenous (reject the H_0: the variable is exogenous), so we need to instrument *Number of Crimes* in a 2SLS framework. On the other hand, if the coefficient on the added regressor is statistically not different from zero, *Number of Crimes* is shown to be exogenous and an OLS using this variable is consistent and 2SLS is not necessary.

We construct *Number of Crimes* variables in multiple forms: *Number of Crimes* (crime per capita for total or by crime type: property, violent or social) and Δ *Crimes* (change in the number of crimes, total or by crime type), as defined in Appendix 1 Table 10. We also test the exogeneity of another key variable, *Distance to Crime Hotspot*.[45]

Empirical Modeling

We perform our main analysis in two steps: (i) traditional ordinary least squares (OLS) and quantile hedonic models, and (ii) two-stage least squares (2SLS) hedonic models and instrumental variable quantile hedonic regressions (IQHR) to account for spatial autocorrelation.

For the first approach, we include demographic and crime variables in a traditional hedonic model to examine their marginal impact on house prices. We estimate our first model using OLS as:

$$\log(Sale\ price) = \beta_0 + \beta X + \phi Demographic\ Variables + \gamma Crime\ Variables + Dummies + \epsilon, \tag{1}$$

where the dependent variable is the log of house sale price; X is a vector of property and neighborhood characteristics; *Demographic Variables* is a vector of demographic variables; *Crime Variables* is a vector of crime variables: number of crimes, change in crime numbers and distance to crime hotspot; *Dummies* are dummies for month and year. As described in "Endogeneity" section, if number of crimes (levels or changes) turns out to be endogenous based on the Wu-Hausman test, we use *911 Calls* as an instrumental variable to perform 2SLS regressions instead of OLS. Precise definitions of the dependent and explanatory variables are provided in Appendix 1 Table 10. For the distance variable, we use the straight (Euclidean) distance to the crime hotspot in our main analysis. Other distance measures (travel miles and travel minutes) are used for robustness checks.

Next, we use a quantile regression approach to examine whether our key explanatory variables have a different impact in the tails of the dependent variable's distribution than in the center of the distribution. Specifically, we estimate the following quantile regressions:

$$\log(Sale\ price) = \beta_0(\tau) + \beta(\tau) X + \phi(\tau) Demographic\ Variables + \gamma(\tau) Crime\ Variables + Dummies + \epsilon, \tag{2}$$

where the dependent and explanatory variables remain the same as in Eq. 1. We run Eq. 2 for quantiles (τ) of house prices: 0.05 (lowest-end), 0.25, 0.50 (median), 0.75 and 0.95 (highest-end).

[45] In addition to performing a Wu-Hausman test on *Distance to Crime Hotspot*, we follow Ceccato and Wilhelmsson (2020) to include area fixed effects (as reported in Appendix 2 Table 11) to further address the potential endogeneity of the distance to hotspot variable.

We then extend the models in Eq. 1 and Eq. 2 by incorporating the spatial autocorrelation between a house's sale price and the sale prices of its neighboring houses. We follow Zhang (2016) to estimate the following two-stage least squares (2SLS) model that accounts for the correlation between residuals from the hedonic models and the prices of neighboring houses. In the first stage, we regress Wy (as defined in "Spatial Autocorrelation" section) against the spatially lagged, exogenous instruments X (hedonic variables from Eq. 1). We then use the coefficient estimates from the first stage to compute the predicted value for $\widehat{Wy}$. In the second stage, we add $\widehat{Wy}$ to Eq. 1 and Eq. 2. The 2SLS hedonic model can be written as:

$$\log(Sale\ price) = \beta_0 + \rho\widehat{W}_y + \beta X + \phi Demographic\ Variables + \gamma Crime\ Variables + Dummies + \epsilon. \tag{3}$$

We also estimate a quantile regression model with the instrumental variable approach, namely the IQHR model as:

$$\log(Sale\ price) = \beta_0(\tau) + \rho(\tau)\widehat{W}_y + \beta(\tau)X + \phi(\tau)Demographic\ Variables + \gamma(\tau)Crime\ Variables + Dummies + \epsilon. \tag{4}$$

Equations 3 and 4 are analogous to Eqs. 1 and 2, except that we use an instrumental variable approach to account for spatial autocorrelation $\left(\widehat{W}_y\right)$.

Empirical Results

Hotspot and Coldspot Analysis

Table 4 presents the results of our crime hotspot and coldspot analysis using the Getis-Ord statistic as our clustering metric. We identify census tracts with high crime intensity (i.e., hotspots) as follows. First, for each year, we count the total number of reported crimes for each census tract and its neighbor tracts. A neighbor tract j for census tract i is determined based on the queen contiguity: any census tract that shares a border (including a vertex) with tract i is considered a neighbor tract j. We use the total numbers of crimes reported in tract i and its neighbor tracts to compute the G_i^* as defined in "Hotspot and Coldspot Characterization" section. Second, for each year, we identify census tracts with G_i^* greater than the average G_i^* across all the tracts for that year. A census tract with significantly higher (lower) than the average G_i^* in a given year is defined as a tract with high (low) crime intensity, or hotspot (coldspot).

The number of crime hotspots varies over time; on average, there are nearly 13 census tracts (or 9.5% of all census tracts) identified as crime hotspots in Seattle. The number of tracts with high crime intensity gradually increases over time and reaches its peak in 2020 (15 tracts). While the number of low crime intensity tracts also averages 13, it has a slightly decreasing trend. Figure 4 illustrates the locations of crime hot and coldspots as well as census tracts with insignificant crime intensity

in years 2010, 2015 and 2020. Consistent with our earlier observation, we find a concentration of crime around the Westlake Center area consistently throughout 2010, 2015 and 2020. The area with crime intensity further expands in 2020.

Wu-Hausman Test and Correlations

We identify a set of 9 possible instruments. We then regress these potentially endogenous variables on *911 Calls* as well as other exogenous variables on the right-hand side of Eq. 1 and obtain the residuals (v). We include v in Eq. 1 and test if its coefficient is statistically different from zero.

In Table 5 Panel A, the computed Wu-Hausman statistic has p-values < 0.05, meaning that we reject the H_o (the variable is exogenous), for *Number of Crimes – Total*, *Property* and *Social*; for changes: Δ *Crimes – Total* and *Property*. With these factors as a right-hand side variable, an OLS model is biased and hence an instrumental variable approach is necessary. Four variables pass the Wu-Hausman test (*Number of Crimes – Violent*, Δ *Crimes—Violent,* Δ *Crimes—Social* and *Distance to Crime Hotspot*) and can thus be used in an OLS regression.

The crime variables that did not pass the Wu-Hausman test are instrumented by the *911 Calls* variables.[46] To validate that *911 Calls* is a viable instrument, we perform a Montiel—Pfluger test to identify a weak instrument: in both cases the variables pass the test.

Panel B reports correlations between these variables. For levels (changes), we compute the aggregate number of 911 calls and crimes per month for *Number of Crimes – Total*, *Violent*, *Property* and *Social*, and *Distance to Crime Hotspot* (*Straight Miles*, *Travel Miles* and *Travel Time*). The correlations between *Number of Crimes – Total* and subsets are positive, as expected. Interestingly, the correlation of *911 Calls* (both for levels and changes) is strongest for *Violent* (0.465)*,* slightly less positive for *Property* (0.268) and *Social* (0.202).

Our 3 measures of *Distance to Crime Hotspot* variable are only significantly correlated with *Violent* (-0.325). These 3 variables, however, are positively correlated with *911 Calls*, suggesting that the farther away from a crime hotspot, the likelier it is that a 911 call will be made. This can be partially explained by the higher allocation of police resources to areas with the highest incidences of crime. The Center for Evidence-Based Crime Policy (CEBCP) reports that the effort to allocate more resources in crime hotspot areas resulted in reduced numbers of calls for service in the affected areas.[47] Consistent with our intuition, our 3 measures of *Distance to Crime Coldspot* are significantly correlated with *Violent* (0.276) and *Social* (0.519).

[46] *Number of Crimes* and Δ *Crimes* are instrumented by *Number of 911 Calls* and Δ *911 Calls*, respectively.

[47] https://havengear.com/blog/increasing-hot-spots-policing-results-in-lower-crime-rates/.

Table 4 Getis-Ord statistics and crime hot/coldspots

Year	# of crimes per tract		Getis-ord statistics (G*)		Hotspot tracts		Coldspot tracts		Non-hot/cold spot tracts	
	Mean	Median	Mean	Median	*N*	(%)	*N*	(%)	*N*	(%)
2008	460	337	0.775%	0.621%	13	9.63%	9	6.67%	113	83.70%
2009	468	350	0.777%	0.621%	13	9.63%	12	8.89%	110	81.48%
2010	452	339	0.776%	0.640%	14	10.37%	12	8.89%	109	80.74%
2011	422	316	0.776%	0.652%	11	8.15%	10	7.41%	114	84.44%
2012	421	318	0.778%	0.633%	12	8.89%	12	8.89%	111	82.22%
2013	456	332	0.774%	0.653%	13	9.63%	12	8.89%	110	81.48%
2014	500	365	0.778%	0.639%	12	8.89%	11	8.15%	112	82.96%
2015	487	349	0.777%	0.640%	13	9.63%	11	8.15%	111	82.22%
2016	505	379	0.777%	0.639%	11	8.15%	7	5.19%	117	86.67%
2017	522	363	0.778%	0.629%	12	8.89%	10	7.41%	113	83.70%
2018	537	352	0.782%	0.622%	14	10.37%	10	7.41%	111	82.22%
2019	500	356	0.783%	0.626%	14	10.37%	9	6.67%	112	82.96%
2020	530	432	0.771%	0.653%	15	11.11%	8	5.93%	112	82.96%

This table reports Getis-Ord statistics and the number of census tracts identified as crime hot/coldspots in each year during the 2008 – 2020 period. For each census tract *i*, we compute the Getis-Ord (G_i^*) statistic as the weighted-average number of crimes reported in its neighboring census tract *j* ($j = 1, 2, \ldots$), divided by the sum of all crime counts across all the census tracts. A census tract *j* is defined as census tract *i*'s neighbor if it shares a border (including a vertex) with tract *i* based on the queen contiguity. A census tract with G_i^* greater (smaller) than the average G_i^* across all the census tracts in a given year is defined as a hot (cold) spot

2SLS and Quantile Regressions

Results Without Spatial Autocorrelation Adjustments

Table 6 displays the results of running 2SLS and quantile regressions without incorporating adjustments for spatial autocorrelation or *Distance to Crime Coldspot*; when these influences are included we find important changes in our results. All estimates use month and year dummies.[48] We have 61,902 observations using *911 Calls* as an instrument for *Number of Crimes – Total.*

Our 2SLS estimates in model 1 are generally consistent with similar hedonic residential pricing models. We find a positive significant impact from: a larger living area or lot size; a younger home; more bathrooms; homes with a fireplace or view.

[48] In the existing hedonic literature, area-fixed effects are usually included to control for unobserved area-specific characteristics to address an omitted variable problem. Abbott and Klaiber (2011) documents that the omitted variable bias remains even with the use of area-fixed effects, both defined at coarse levels (cities or counties) or at much finer scales (zip areas or census tracts/blocks). Our baseline models (Eqs. 1 and 2) exclude area-fixed effects, and we address the omitted variable issue by including the spatial autocorrelation variable, *Wy* (Eqs. 3 and 4), to capture important neighborhood characteristics shared with nearby residents.

Table 5 Endogeneity, instrumental variables and correlations

Panel A: Tests for endogeneity and weak instrument

	Wu-Hausman F-Stats (*p*-value) (H_0: Variable is exogenous)		Need an IV?
Number of crimes	16.496***	(<0.001)	Yes
Number of crimes—property	22.232***	(<0.001)	Yes
Number of crimes—violent	0.008	(0.928)	No
Number of crimes—social	29.296***	(<0.001)	Yes
Δ crimes	5.627**	(0.018)	Yes
Δ crimes—property	4.320**	(0.038)	Yes
Δ crimes—violent	0.358	(0.550)	No
Δ crimes—social	0.889	(0.342)	No
Distance to crime hotspot	0.725	(0.346)	No

*, **, and *** indicate significance at the 10, 5 and 1 percent levels, respectively

	Montiel—Pflueger effective F-Stats (H_0: Variable is a weak instrument)	Critical value (tau = 5%)	Weak IV?
Number of 911 calls	3,374.996	37.418	No
Δ 911 calls	374.025	37.418	No

Table 5 (continued)

Panel B: Correlations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1) Number of 911 Calls	1.000															
(2) Number of Crimes—Total	0.344*	1.000														
(3) Number of Crimes—Violent	0.465*	0.497*	1.000													
(4) Number of Crimes—Property	0.268*	0.964*	0.301*	1.000												
(5) Number of Crimes—Social	0.202*	0.288*	0.478*	0.050	1.000											
(6) Δ 911 Calls	0.245*	0.183*	0.158	0.159	0.088	1.000										
(7) Δ Crimes—Total	0.141	0.595*	0.139	0.601*	0.153	0.481*	1.000									
(8) Δ Crimes—Violent	0.154	0.218*	0.319*	0.166	0.128	0.542*	0.448*	1.000								
(9) Δ Crimes—Property	0.130	0.606*	0.120	0.622*	0.120	0.441*	0.991*	0.368*	1.000							
(10) Δ Crimes—Social	0.117	0.135	0.046	0.086	0.286*	0.362*	0.533*	0.292*	0.446*	1.000						
(11) Distance to Crime Hotspot (Straight Miles)	0.410*	−0.067	−0.325*	0.012	−0.186*	0.027	0.054	0.007	0.049	0.074	1.000					
(12) Distance to Crime Hotspot (Travel Miles)	0.423*	−0.010	−0.252*	0.072	−0.234*	0.013	0.061	0.004	0.057	0.083	0.966*	1.000				
(13) Distance to Crime Hotspot (Travel Time)	0.428*	−0.024	−0.317*	0.092	−0.363*	0.033	0.054	0.003	0.053	0.054	0.902*	0.948*	1.000			
(14) Distance to Crime Coldspot (Straight Miles)	0.054	0.005	0.276*	−0.137	0.519*	0.021	−0.001	0.049	−0.009	0.029	−0.108	−0.152	−0.255*	1.000		
(15) Distance to Crime Coldspot (Travel Miles)	0.098	0.019	0.348*	−0.141	0.564*	0.022	0.000	0.054	−0.008	0.023	−0.144	−0.191*	−0.291*	0.957*	1.000	
(16) Distance to Crime Coldspot (Travel Time)	0.231*	0.122	0.480*	−0.051	0.580*	0.019	0.007	0.059	0.001	0.018	−0.193*	−0.217*	−0.311*	0.900*	0.951*	1.000

Panel A reports the results of the Wu-Hausman test for endogeneity of the crime measure variables (as defined in Appendix 1 Table 10) and the result of the Montiel – Pflueger weak instrument test for *Number of 911 Calls* and *Δ911 Calls*. The Wu-Hausman test is performed based on Eq. 1. Panel B reports the correlations between the instrumental variables and the crime measure variables

* shows significance at the 0.05 level

Table 6 2SLS and quantile regressions – results without spatial autocorrelation adjustments

Using *Number of 911 Calls* as an IV for *Number of Crimes*						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Living area	0.320***	0.286***	0.307***	0.321***	0.341***	0.348***
	(0.003)	(0.006)	(0.003)	(0.003)	(0.004)	(0.005)
Lot area	0.021***	0.008***	0.013***	0.020***	0.026***	0.038***
	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Bedrooms	−0.019***	−0.009**	−0.012***	−0.012***	−0.014***	−0.018***
	(0.002)	(0.003)	(0.002)	(0.001)	(0.001)	(0.003)
Bathrooms (Full)	0.003	−0.010**	0.002	0.005***	0.004**	0.010***
	(0.002)	(0.004)	(0.003)	(0.002)	(0.002)	(0.003)
Age	−0.000***	−0.001***	−0.000***	−0.000***	−0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.093***	−0.062***	−0.076***	−0.081***	−0.087***	−0.099***
	(0.007)	(0.005)	(0.002)	(0.003)	(0.002)	(0.003)
Renovated	−0.046***	−0.275***	−0.107***	−0.006	0.025***	0.047***
	(0.011)	(0.017)	(0.012)	(0.006)	(0.005)	(0.011)
Fireplace	0.040***	0.062***	0.041***	0.029***	0.020***	0.014***
	(0.005)	(0.008)	(0.002)	(0.001)	(0.002)	(0.003)
Lake/Mountain view	0.105***	0.065***	0.069***	0.081***	0.104***	0.176***
	(0.006)	(0.007)	(0.004)	(0.004)	(0.006)	(0.008)
Population	0.012**	−0.027***	0.000	0.014***	0.022***	0.027***
	(0.005)	(0.008)	(0.006)	(0.004)	(0.004)	(0.007)
% Male population	0.726***	1.706***	1.043***	0.762***	0.521***	1.083***
	(0.153)	(0.206)	(0.100)	(0.093)	(0.101)	(0.237)
% White population	0.793***	1.093***	0.792***	0.780***	0.773***	0.749***
	(0.034)	(0.163)	(0.022)	(0.024)	(0.019)	(0.026)
% Asian population	0.043	0.080	0.011	0.088**	0.160***	0.230***
	(0.032)	(0.149)	(0.023)	(0.035)	(0.022)	(0.053)
% Old population	0.283***	0.427***	0.124***	0.080***	0.078**	0.372***
	(0.042)	(0.107)	(0.040)	(0.029)	(0.031)	(0.057)
Number of crimes (IV)	−0.828**	−0.117	−0.948***	−0.798***	−0.757***	−1.104***
	(0.347)	(0.790)	(0.170)	(0.176)	(0.136)	(0.183)
Distance to crime hotspot	−0.072***	−0.072***	−0.068***	−0.070***	−0.073***	−0.084***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	12.295***	11.659***	12.199***	12.274***	12.385***	12.216***
	(0.078)	(0.150)	(0.055)	(0.033)	(0.047)	(0.088)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902

This table reports the results of 2SLS regression (model 1, as estimated in Eq. 1) and quantile regressions (models 2 – 6, as estimated in Eq. 2) for 5, 25, 50, 75 and 95 percent quantiles, respectively. Definitions of the variables are provided in Appendix 1 Table 10. *Number of crimes* is instrumented by *Number of 911 calls*. Standard errors are reported in parentheses: for the 2SLS regression (model 1), standard errors are clustered by year and month. For quantile regressions (models 2 – 6), standard errors are obtained through bootstrap replications. *, ** and *** denote significance at the 10, 5 and 1 percent levels, respectively

Additional bedrooms and houses with a single floor exhibit a negative relation, since living area is already factored into the model. In terms of demographics, the 2SLS estimates are significantly positive for *% White, % Male* and *% Old Population.*

The quantile regressions in models 2 to 6 allow us to examine how these 2SLS results vary across the price quantiles: 0.05 (low end), 0.25, 0.50 (median), 0.75 and 0.95 (high end). The most evident trends are increasing significance (from low to high prices) of: lot area and view and the decreasing significance of fireplace. The sign on *% Asian Population* is positive in the higher quantiles.When we include zip code fixed effects and re-run these regressions (results are reported in Appendix 2 Table 11), the sign on *% Asian Population* becomes significant and negative in all quantiles.[49] The coefficients on other demographic variables generally retain the same sign with zip code fixed effects.

Our focus however is on the impact of crime and we see a significant negative relation between *911 Calls* and housing prices in each of the quantile regressions, except for the lowest priced houses, where there is no significant relation. This fact is consistent with the notion that increasing crime rates (as instrumented by *911 Calls*) depress housing prices. We find a significant negative relation between the distance to a crime hotspot and housing prices. The latter, perhaps surprising result, is discussed further below.

We re-estimate this regression using different crime variables.[50] When we substitute *Number of Crimes—Violent* for *Number of 911 Calls*, its coefficient is significantly negative across all the quantiles. The coefficients on the demographic variables generally retain the same signs and approximately the same magnitudes as in Table 6.

We next use Δ *Violent Crime* as a crime rate proxy and employ changes in demographic variables.[51] The coefficients on the demographic variable changes again generally retain the same sign as in Table 6, except for the Δ *Old Population* variable now has a significant negative sign, suggesting that an increasingly old population depresses prices. The coefficient on Δ *Violent Crime* is significantly positive in all but the median quantile. The negative estimates for coefficient on distance to crime hotspot are virtually identical to those obtained previously.

Next we employ Δ *Social Crime*. The coefficients are positive and significant in each quartile. In this and in the previous estimate, we find a strong negative impact from increasing population. This effect is dampened when we refine our estimation in the following tables. Furthermore, our discussions with Port of Seattle police have

[49] The inclusion of zip code fixed effects accounts for certain unobserved neighborhood characteristics, one of which could be gentrification. Hwang (2020) studies Seattle's gentrification from the 1970's through 2010's, and shows that neighborhoods with a higher Asian population experienced less gentrification, while those with a higher Black population were gentrified more frequently. She explains that newly arriving immigrants in Seattle, who are predominantly Asian recently, tend to settle in areas with an existing significant Asian population. This influx can deter gentrification in such areas, potentially leading to lower housing prices (compared to areas which recently have been gentrified).

[50] Here we only report the most relevant results for these alternative measures. The complete regressions are available from the authors.

[51] When we use changes in proxies for crime rates, we also use changes in demographic variables.

indicated that under-reporting of social crimes has had a increasing trend recently, as the Seattle Police Department experienced a significant decrease in staffing from the late 2010s.[52]

Lastly, we use lagged *911 Calls* as our proxy for crime rates. Its impact is significantly negative for each quantile, except for the lowest priced residences, while *Distance to Crime Hotspot* has a relatively consistent negative impact. The R^2 for these 5 regressions are all approximately 77%.

We can interpret the key results as follows (for the median quantile estimates for simplicity): a 1 percentage point increase in the *Number of Crimes* (crime per capita, instrumented by 911 calls) leads to a 0.550% decrease in prices; a 1 percentage point increase in the *Number of Crimes – Violent* (crime per capita – violent only) leads to a 0.992% decrease in prices; a 1% increase in the number of violent crimes has an insignificant effect; a 1% increase in the number of social crimes implies a 1.92% increase in prices; a 1 percentage point increase in the *Number of Crimes – Lag* 1 (instrumented by 911 calls, lagged one year) leads to a 0.505% decrease in prices. So *Number of 911 Calls* (contemporaneous or lagged) has the expected negative impact on pricing.

If the distance to the hotspot increases by one mile, prices *decrease* by an average of 7.08%. In other words, the closer the property is located to the crime hotspot, the more expensive the sale price. While this result may be *prima facie* counterintuitive, as we observe in Fig. 4, crime hotspots (defined by the Getis-Ord statistics) cluster around the city center area, covering the Westlake Center, Pioneer Square, Central Business District and University District. From Fig. 2, we see that high-end houses (the darkest brown dots) are concentrated around the north (and northeast) of the city center area. We may conjecture that buyers are willing to pay higher prices for closer proximity to the city center (e.g., for easier access to amenities).[53] The previous literature also documents that distance to the city center has a significant negative impact on prices.

Results with Spatial Autocorrelation Adjustments

However, these results are somewhat suspect because they fail to account for spatial autocorrelation. The estimates using the *Wy* matrix approach, described in "Spatial Autocorrelation" section, are presented in Table 7. Panels A to E use the same explanatory variables (together with *Wy*) as in Table 6 and the associated discussion. All estimates use month and year dummies. We have 61,902 observations in Panels A through D; there are 58,301 observations in Panel E. For brevity we only discuss results that differ significantly from those in Table 6.

The first important observation is that in all 5 panels of Table 7 the spatial lag coefficient is positive and significant across each of the price quantiles: in Panel A

[52] According to the most recent FBI data, 2019, the average over the 50 largest jurisdictions is 26.9 officers for every 10,000 residents. The corresponding figure for Seattle is 18.5 per 10,000.

[53] Further justification for this observation is by noting the 0.95 correlation between distance to CBD and distance to crime hotspot, which is consistent with our hypothesis that criminals are drawn to locales that offer greater possible gains.

the coefficients range from 0.364 to 0.414 with an average value of 0.377; the other panels show very similar magnitudes. This indicates statistically significant positive spatial autocorrelation, as expected.[54]

In Panel A the *911 Calls* variable is positive and significant for all but the highest priced residences.[55] The distance to a crime hotspot also remains negative and significant. In Panel B the estimates for *Number of Crimes – Violent* are significantly negative in each quantile. The estimates generally decrease in magnitude as the house price increases. The estimates for *Distance to Crime Hotspot* remain negative and significant. The R^2 for these regressions are again very similar, but somewhat higher than in Table 6, averaging 82%, which implies that the adjustment for spatial autocorrelation results in an improved fit.

The interpretation of the estimated coefficients incorporating spatial correlation is somewhat different from the usual OLS approach.[56] For example, the coefficient for 911 calls at the median is 0.373, which represents the direct impact of the variable; with the spatial autocorrelation coefficient ρ of 0.364, the total impact[57] is calculated as $0.373 \times (1-\rho)^{-1} = \frac{0.373}{1-0.364} = 0.586$. Moreover, the coefficient for *Distance to Crime Hotspot* at the median is -0.040, the direct impact of the variable; by incorporating spatial autocorrelation, the total impact is $\frac{-0.040}{1-0.364} = -0.063$. This figure is similar to the -0.070 number estimated in Table 6 without spatial correlation adjustments.

We re-estimated these regressions using as a crime rate proxy Δ *Violent Crime* and Δ *Social Crime* (in Panels C and D, respectively). In the first case the estimated coefficients were generally insignificant; in the second case they were positive and significant for the 5%, 50% and 75% quantiles. In both cases the *Distance to Crime Hotspot* variable was consistently negative and significant. Panel E, using lagged *911 Calls*, shows estimates very similar in sign and magnitude to those obtained in Panel A.

Overall the results from Tables 6 and 7 indicate that *Distance to Crime Hotspot* has a consistently negative impact on prices; the same result holds for *Number of Crimes – Violent*. Δ *Violent Crime* has a significant positive effect, but this relation disappears when we incorporate spatial correlation in lower priced residences. Δ *Social Crime* has a significant positive effect.

[54] These *Wy* estimates are much higher than those estimated in Zhang (2016). A partial explanation for this is the much higher population density in our sample (Seattle versus the Fargo-Moorehead MSA).

[55] Chamberlain and Boggess (2016) corroborates this result: they examine how burglars decide which neighborhoods to target, whether rich or poor. They show that, despite higher potential gains by burgling in wealthy neighborhoods, burglars often opt for familiar, disadvantaged neighborhoods close to their own residences. Thus, neighborhoods with the highest priced houses are least likely to be targeted by burglars, leading to much lower frequencies of 911 calls.

[56] See Pace and LeSage (2009).

[57] The spatial-lag matrix, *Wy*, is standardized, such that its row elements sum to one. See Zhang (2016) and also Kim et al. (2003) for technical details on the use of spatial weight matrix in an instrumental variable framework.

Table 7 2SLS and instrumental quantile regressions – results with spatial autocorrelation adjustments

Panel A: Using *Number of 911 Calls* as an IV for *Number of Crimes*						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.394***	0.414***	0.368***	0.364***	0.364***	0.376***
	(0.013)	(0.017)	(0.006)	(0.004)	(0.008)	(0.013)
Living area	0.261***	0.241***	0.252***	0.267***	0.286***	0.299***
	(0.004)	(0.006)	(0.003)	(0.003)	(0.002)	(0.005)
Lot area	0.018***	0.008***	0.014***	0.018***	0.022***	0.034***
	(0.001)	(0.002)	(0.000)	(0.001)	(0.001)	(0.001)
Bedrooms	−0.007***	−0.000	−0.002*	−0.002*	−0.002*	−0.008**
	(0.001)	(0.004)	(0.001)	(0.001)	(0.001)	(0.004)
Bathrooms (Full)	0.008***	−0.007***	0.006***	0.009***	0.009***	0.015***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Age	−0.001***	−0.002***	−0.001***	−0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.076***	−0.053***	−0.064***	−0.067***	−0.071***	−0.084***
	(0.005)	(0.007)	(0.002)	(0.002)	(0.002)	(0.004)
Renovated	−0.028***	−0.264***	−0.102***	0.021***	0.048***	0.069***
	(0.011)	(0.012)	(0.012)	(0.006)	(0.002)	(0.010)
Fireplace	0.025***	0.044***	0.025***	0.013***	0.010***	0.007***
	(0.004)	(0.007)	(0.002)	(0.002)	(0.002)	(0.002)
Lake/Mountain view	0.063***	0.037***	0.036***	0.047***	0.059***	0.107***
	(0.005)	(0.006)	(0.003)	(0.003)	(0.003)	(0.004)
Population	−0.001	−0.017	−0.005	0.001	0.003	0.013**
	(0.002)	(0.013)	(0.003)	(0.004)	(0.004)	(0.005)
% Male population	−0.188*	−0.241	0.070	0.052	0.009	0.377***
	(0.099)	(0.234)	(0.060)	(0.079)	(0.074)	(0.142)
% White population	0.605***	0.883***	0.626***	0.557***	0.520***	0.458***
	(0.029)	(0.067)	(0.018)	(0.015)	(0.018)	(0.020)
% Asian population	0.147***	0.234**	0.102***	0.129***	0.152***	0.151***
	(0.016)	(0.104)	(0.031)	(0.027)	(0.019)	(0.032)
% Old population	−0.167***	−0.246***	−0.184***	−0.206***	−0.227***	−0.222***
	(0.028)	(0.053)	(0.030)	(0.025)	(0.015)	(0.032)
Number of crimes (IV)	0.730***	1.294***	0.599***	0.373***	0.214*	−0.111
	(0.148)	(0.310)	(0.094)	(0.100)	(0.118)	(0.126)
Distance to crime hotspot	−0.038***	−0.036***	−0.038***	−0.040***	−0.042***	−0.046***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	7.511***	6.972***	7.715***	7.855***	7.932***	7.670***
	(0.180)	(0.221)	(0.076)	(0.078)	(0.076)	(0.181)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902
Panel B: Using *Number of Crimes—Violent Only*						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.376***	0.371***	0.348***	0.350***	0.354***	0.372***
	(0.012)	(0.016)	(0.006)	(0.003)	(0.007)	(0.012)

Table 7 (continued)

Living area	0.260***	0.237***	0.252***	0.266***	0.285***	0.299***
	(0.004)	(0.005)	(0.003)	(0.002)	(0.002)	(0.005)
Lot area	0.018***	0.005***	0.013***	0.017***	0.022***	0.034***
	(0.001)	(0.002)	(0.000)	(0.001)	(0.001)	(0.001)
Bedrooms	−0.008***	−0.003	−0.003*	−0.003**	−0.003**	−0.008**
	(0.001)	(0.004)	(0.001)	(0.001)	(0.001)	(0.004)
Bathrooms (Full)	0.008***	−0.006***	0.005*	0.009***	0.009***	0.014***
	(0.001)	(0.002)	(0.003)	(0.001)	(0.002)	(0.002)
Age	−0.001***	−0.002***	−0.001***	−0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.080***	−0.061***	−0.068***	−0.070***	−0.072***	−0.085***
	(0.005)	(0.005)	(0.002)	(0.003)	(0.003)	(0.003)
Renovated	−0.028***	−0.262***	−0.104***	0.021***	0.048***	0.070***
	(0.011)	(0.012)	(0.017)	(0.006)	(0.003)	(0.009)
Fireplace	0.025***	0.047***	0.026***	0.014***	0.010***	0.006**
	(0.004)	(0.005)	(0.002)	(0.002)	(0.002)	(0.003)
Lake/Mountain view	0.063***	0.036***	0.037***	0.046***	0.060***	0.107***
	(0.005)	(0.005)	(0.002)	(0.004)	(0.003)	(0.004)
Population	0.012***	0.011	0.012***	0.015***	0.009***	0.015***
	(0.002)	(0.009)	(0.002)	(0.003)	(0.003)	(0.005)
% Male population	0.307***	0.787***	0.525***	0.440***	0.274***	0.465***
	(0.070)	(0.128)	(0.041)	(0.047)	(0.062)	(0.106)
% White population	0.455***	0.602***	0.476***	0.454***	0.449***	0.410***
	(0.023)	(0.037)	(0.017)	(0.012)	(0.014)	(0.021)
% Asian population	0.006	0.021	−0.026	0.041*	0.078***	0.097**
	(0.028)	(0.053)	(0.029)	(0.022)	(0.018)	(0.039)
% Old population	−0.183***	−0.172***	−0.212***	−0.223***	−0.222***	−0.214***
	(0.022)	(0.044)	(0.027)	(0.022)	(0.015)	(0.026)
Number of crimes—violent	−2.139***	−3.152***	−2.447***	−2.130***	−1.472***	−1.577***
	(0.455)	(0.578)	(0.322)	(0.211)	(0.151)	(0.331)
Distance to crime hotspot	−0.040***	−0.040***	−0.039***	−0.041***	−0.043***	−0.046***
	(0.002)	(0.002)	(0.001)	(0.000)	(0.001)	(0.001)
Constant	7.622***	7.208***	7.827***	7.884***	7.981***	7.709***
	(0.177)	(0.196)	(0.076)	(0.069)	(0.065)	(0.168)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902
Panel C: Using *Δ Crimes—Violent*						
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.425***	0.431***	0.395***	0.388***	0.398***	0.413***
	(0.018)	(0.017)	(0.005)	(0.006)	(0.011)	(0.014)
Living area	0.268***	0.252***	0.259***	0.276***	0.288***	0.302***
	(0.003)	(0.006)	(0.002)	(0.003)	(0.002)	(0.005)
Lot area	0.013***	−0.000	0.009***	0.012***	0.018***	0.030***
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
Bedrooms	−0.011***	−0.009**	−0.007***	−0.006***	−0.004***	−0.008**
	(0.002)	(0.004)	(0.001)	(0.001)	(0.001)	(0.004)

Table 7 (continued)

Bathrooms (Full)	0.005***	−0.012**	0.000	0.005**	0.008***	0.016***
	(0.001)	(0.005)	(0.001)	(0.002)	(0.001)	(0.003)
Age	−0.001***	−0.001***	−0.001***	−0.000***	−0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.080***	−0.066***	−0.068***	−0.067***	−0.071***	−0.085***
	(0.005)	(0.005)	(0.002)	(0.002)	(0.003)	(0.004)
Renovated	−0.031***	−0.259***	−0.103***	0.015*	0.049***	0.065***
	(0.011)	(0.010)	(0.012)	(0.008)	(0.003)	(0.011)
Fireplace	0.031***	0.061***	0.031***	0.016***	0.014***	0.010***
	(0.005)	(0.006)	(0.003)	(0.002)	(0.002)	(0.003)
Lake/Mountain view	0.060***	0.039***	0.040***	0.045***	0.053***	0.093***
	(0.005)	(0.004)	(0.003)	(0.003)	(0.003)	(0.005)
Δ Population	−6.758***	−7.272***	−8.306***	−6.953***	−6.174***	−7.300***
	(1.261)	(1.100)	(0.488)	(0.637)	(0.737)	(0.742)
Δ Male population	0.541	−0.057	1.574**	0.607	0.403	2.223***
	(1.205)	(1.394)	(0.632)	(0.503)	(0.497)	(0.691)
Δ White population	3.432***	3.445***	3.560***	3.744***	3.715***	3.541***
	(0.952)	(0.238)	(0.218)	(0.307)	(0.263)	(0.282)
Δ Asian population	3.347***	4.176***	3.646***	3.242***	2.895***	2.525***
	(0.259)	(0.234)	(0.085)	(0.062)	(0.076)	(0.141)
Δ Old population	−0.652**	−0.968***	−0.867***	−0.741***	−0.581***	−0.404***
	(0.301)	(0.251)	(0.095)	(0.091)	(0.058)	(0.098)
Δ Crimes—violent	−0.002	0.002	0.002	0.000	−0.000	0.005*
	(0.009)	(0.011)	(0.002)	(0.004)	(0.002)	(0.003)
Distance to crime hotspot	−0.031***	−0.028***	−0.032***	−0.032***	−0.034***	−0.037***
	(0.004)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	7.352***	7.077***	7.708***	7.857***	7.783***	7.663***
	(0.246)	(0.204)	(0.079)	(0.082)	(0.146)	(0.180)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902
Panel D: Using *Δ Crimes—Social*						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.424***	0.429***	0.395***	0.388***	0.397***	0.413***
	(0.017)	(0.016)	(0.005)	(0.006)	(0.011)	(0.014)
Living area	0.268***	0.252***	0.259***	0.276***	0.288***	0.302***
	(0.003)	(0.005)	(0.002)	(0.002)	(0.002)	(0.005)
Lot area	0.013***	−0.001	0.009***	0.012***	0.018***	0.030***
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
Bedrooms	−0.011***	−0.009**	−0.007***	−0.006***	−0.004***	−0.008**
	(0.002)	(0.004)	(0.001)	(0.001)	(0.001)	(0.004)
Bathrooms (Full)	0.005***	−0.012**	0.000	0.005**	0.008***	0.016***
	(0.001)	(0.005)	(0.001)	(0.002)	(0.001)	(0.003)
Age	−0.001***	−0.001***	−0.001***	−0.000***	−0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.080***	−0.066***	−0.068***	−0.067***	−0.071***	−0.085***
	(0.005)	(0.005)	(0.002)	(0.003)	(0.003)	(0.004)

Table 7 (continued)

Renovated	−0.031***	−0.260***	−0.103***	0.014*	0.049***	0.066***
	(0.011)	(0.011)	(0.012)	(0.007)	(0.003)	(0.012)
Fireplace	0.031***	0.061***	0.031***	0.016***	0.014***	0.010***
	(0.005)	(0.006)	(0.003)	(0.002)	(0.002)	(0.003)
Lake/Mountain view	0.060***	0.040***	0.040***	0.045***	0.054***	0.093***
	(0.005)	(0.005)	(0.003)	(0.003)	(0.003)	(0.005)
Δ Population	−6.743***	−7.292***	−8.304***	−6.972***	−6.166***	−7.275***
	(1.257)	(1.092)	(0.527)	(0.616)	(0.733)	(0.749)
Δ Male population	0.513	−0.079	1.570**	0.618	0.370	2.208***
	(1.207)	(1.382)	(0.626)	(0.532)	(0.539)	(0.674)
Δ White population	3.440***	3.465***	3.560***	3.745***	3.724***	3.545***
	(0.950)	(0.215)	(0.215)	(0.310)	(0.259)	(0.290)
Δ Asian population	3.347***	4.180***	3.645***	3.239***	2.901***	2.519***
	(0.260)	(0.219)	(0.093)	(0.057)	(0.075)	(0.139)
Δ Old population	−0.652**	−0.982***	−0.866***	−0.732***	−0.574***	−0.406***
	(0.299)	(0.243)	(0.088)	(0.092)	(0.062)	(0.096)
Δ Crimes—social	0.005	0.007**	0.002	0.006***	0.006***	0.004
	(0.006)	(0.003)	(0.002)	(0.002)	(0.002)	(0.004)
Distance to crime hotspot	−0.031***	−0.029***	−0.032***	−0.032***	−0.034***	−0.037***
	(0.004)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	7.361***	7.107***	7.710***	7.863***	7.788***	7.665***
	(0.243)	(0.201)	(0.079)	(0.085)	(0.145)	(0.175)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902
Panel E: Using *911 Calls (Lag 1) as an IV for Number of Crimes (Lag 1)*						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.398***	0.428***	0.370***	0.365***	0.363***	0.377***
	(0.015)	(0.010)	(0.008)	(0.005)	(0.004)	(0.007)
Living area	0.262***	0.242***	0.253***	0.267***	0.287***	0.300***
	(0.004)	(0.006)	(0.002)	(0.003)	(0.004)	(0.003)
Lot area	0.018***	0.009***	0.014***	0.018***	0.022***	0.034***
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
Bedrooms	−0.007***	0.001	−0.001	−0.001	−0.002	−0.008**
	(0.001)	(0.003)	(0.002)	(0.002)	(0.003)	(0.004)
Bathrooms (Full)	0.009***	−0.005	0.007***	0.009***	0.009***	0.015***
	(0.001)	(0.006)	(0.001)	(0.002)	(0.001)	(0.001)
Age	−0.001***	−0.002***	−0.001***	−0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.075***	−0.050***	−0.064***	−0.066***	−0.071***	−0.083***
	(0.005)	(0.005)	(0.004)	(0.003)	(0.003)	(0.004)
Renovated	−0.027**	−0.265***	−0.102***	0.022***	0.049***	0.072***
	(0.011)	(0.015)	(0.015)	(0.006)	(0.004)	(0.014)
Fireplace	0.025***	0.043***	0.024***	0.014***	0.011***	0.006*
	(0.004)	(0.006)	(0.002)	(0.003)	(0.002)	(0.003)
Lake/Mountain view	0.062***	0.038***	0.035***	0.047***	0.059***	0.103***
	(0.005)	(0.008)	(0.005)	(0.004)	(0.005)	(0.012)

Table 7 (continued)

Population	−0.006**	−0.026**	−0.009**	−0.002	0.000	0.012***
	(0.003)	(0.011)	(0.004)	(0.003)	(0.003)	(0.004)
% Male population	−0.339**	−0.674**	−0.018	−0.057	−0.070	0.316**
	(0.142)	(0.316)	(0.131)	(0.114)	(0.126)	(0.148)
% White population	0.636***	0.922***	0.639***	0.578***	0.540***	0.467***
	(0.038)	(0.063)	(0.016)	(0.016)	(0.009)	(0.016)
% Asian population	0.174***	0.220**	0.105***	0.147***	0.169***	0.167***
	(0.023)	(0.104)	(0.026)	(0.026)	(0.018)	(0.028)
% Old population	−0.155***	−0.338***	−0.176***	−0.208***	−0.223***	−0.236***
	(0.031)	(0.091)	(0.032)	(0.030)	(0.031)	(0.051)
Number of crimes (IV—Lag1)	1.106***	1.773***	0.784***	0.575***	0.429***	0.019
	(0.223)	(0.403)	(0.130)	(0.095)	(0.139)	(0.197)
Distance to crime hotspot	−0.038***	−0.035***	−0.038***	−0.040***	−0.042***	−0.046***
	(0.002)	(0.002)	(0.000)	(0.001)	(0.001)	(0.002)
Constant	7.522***	7.001***	7.739***	7.880***	7.973***	7.670***
	(0.189)	(0.162)	(0.051)	(0.050)	(0.062)	(0.105)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	58,301	58,301	58,301	58,301	58,301	58,301

This table reports the results of 2SLS regression (model 1, as estimated in Eq. 3) and instrumental quantile regressions (models 2 – 6, as estimated in Eq. 4) for 5, 25, 50, 75 and 95 percent quantiles, respectively. Definitions of the variables are provided in Appendix 1 Table 10. Panels A and B use *Number of crimes* (instrumented by *Number of 911 calls*) and *Number of crimes-violent)*, respectively, as well as levels of the demographic variables (*Population, % Male, % White, % Asian and % Old population*) as control variables. Panels C and D use annual changes for *Number of crimes- violent* and *Number of crimes-social*, respectively, as well as for the demographic variables. Panel E uses lagged values of *Number of crimes* (instrumented by lagged *Number of 911 calls*). Standard errors are reported in parentheses: for 2SLS regressions (model 1), standard errors are clustered by year and month. For quantile regressions (models 2 – 6), standard errors are obtained through bootstrap replications. *, ** and *** denote significance at the 10, 5 and 1 percent levels, respectively

When we focus on 911 calls the situation is less uniform. In Table 6 the coefficient is negative and significant, suggesting that a higher frequency of 911 calls depresses prices. However when we incorporate spatial correlation the sign becomes positive and significant. Our estimates are similar when we use lagged 911 calls by one year. These results suggest that more 911 calls are made in an area with higher house prices, once we incorporate spatial autocorrelation. One potential explanation is that *Wy* captures characteristics specific to certain neighborhoods that are not observable to econometricians, such as "collective efficacy.[58]" Once such characteristics are controlled for, an increase in the number of 911 calls can indicate the residents' greater efforts to monitor the safety of their residence, to report any suspicious persons/activities in the area, etc. A study by Vera Institute of Justice[59]

[58] Collective efficacy is defined in Sampson et al. (1997) as (from the abstract): *social cohesion among neighbors combined with their willingness to intervene on behalf of the common good.* They show that neighborhoods with high collective efficacy tend to experience lower rates of violent crime.

[59] https://www.vera.org/publications/making-911-accessible

corroborates this perspective: among different types of 911 calls made in New Orleans, they report that 911 calls made for property crimes and those for "suspicious behaviors" are the two predominant call types, followed by domestic violence, violent crime, drugs and sex offenses. This trend is observed in both census tracts with the lowest and highest 10 percent median income. Thus, the positive relation between the frequency of 911 calls and housing prices (once we incorporate spatial autocorrelation) can be indicative of residents' collective efficacy or criminals' attempts to target higher priced properties.

Robustness

Our first robustness check restricts our *911 Calls* to only those with high priority. Each 911 call is assigned a priority level, ranging from 1 (highest) to 9 (lowest), based on the severity of an event. *911 Calls (High Priority)* are defined as those with the two highest priority levels. In our data, this subsample represents 68.1% of the 1,528,303 calls in our total sample. By excluding calls of less severity, this measure of crime levels can hopefully become more accurate.

Table 8 Panel A presents our basic regression without adjustment for spatial autocorrelation. Here the coefficient on *911 Calls* (*High Priority)* is negative and significant in each quantile (except for the lowest quantile), with higher values than those reported in the previous tables, suggesting that this measure is a more efficient proxy for crime rates than the total of *911 Calls*. Panel B now include the *Wy* variable. As in Tables 6 and 7, we now observe positive and significant estimates for *911 Calls* (*High Priority)* in the lowest 3 price quantiles. This flip from negative to positive signs on the crime proxies when we include spatial autocorrelation parallels our earlier results. The positive sign is more consistent with our estimates involving distance to crime hotspots, which could suggest that ignoring spatial autocorrelation can lead to spurious estimates. Very similar results obtain when we substitute lagged values of *911 Calls* (*High Priority)* for results with and without spatial autocorrelation adjustments.

Our next robustness check is presented in Table 9, which adds distance to crime coldspot (*Distance to Crime Coldspot*) to the estimation. The rationale is that consumers should be willing to pay more for residences near low crime areas. Table 9 Panel A presents the results using 911 calls as an instrument without correcting for spatial correlation. The estimates are very similar to those reported in Table 6 with the additional observation that *Distance to Crime Coldspot* has a weakly negative impact; its coefficient is small but significant. When we repeat this analysis using *Number of Crimes – Violent,* Δ *Violent Crime,* Δ *Social Crime* and *911 Calls (Lag 1),* we generate very similar estimates for the *Distance to Crime Coldspot* variable.

Table 9 Panel B now incorporates the *Wy* variable and uses *911 Calls* as the proxy for crime rates. We observe the same pattern as in Panel A: the estimates on *Distance to Crime Coldspot* are negative and significant in each quantile, however the average coefficient is only 45% of the value obtained in Panel A. Overall these estimates provide consistent evidence of a small positive pricing impact of being closer to a crime coldspot. When we re-estimate this regression using the other 4 crime rate proxies the results are qualitatively similar.

Conclusions

This study explores the link between crime rates and the pricing of residential housing using a rich data set including reported crimes, 911 calls, housing prices and characteristics, as well as demographic data for the City of Seattle from January 1, 2008 to December 31, 2020. We mitigate the problem of the endogeneity between crime rates and housing prices principally by using the number of 911 calls per police beat as an instrument. We correct for spatial autocorrelation to account for the positive correlations between hedonic model pricing errors for nearby houses. We then use quantile regression to evaluate how these model factor estimates vary across different price strata.

We have three major findings related to our first hypothesis:

(i) The *Distance to Crime Hotspot* variable has a consistently negative and significant coefficient, suggesting that criminals have a tendency to operate in wealthier tracts. Specifically, moving one mile farther from a crime hotspot in a census tract leads to a 7.08% (6.28%) *decrease* in house prices in that tract without (with) adjustment for spatial autocorrelation.

(ii) The impact of crime on housing prices is significant, but of inconsistent sign, in our estimations: a 1 percentage point increase in the number of crimes (crime per capita, instrumented by the number of 911 calls) leads to approximately a 0.55% *decrease* in house prices without adjusting for spatial autocorrelation, but a 0.80% *increase* once we include spatial autocorrelation in the hedonic models. When we restrict the sample to 911 calls with high priority, the corresponding figures are a 0.65% *decrease* and a 0.21% *increase*, respectively. One can speculate that this sign reversal can be attributed to the impact of spatial autocorrelation: once we employ this refinement the result that crime is related to higher price housing is consistent with our estimation using *Distance to Crime Hotspot* and to some of the most recent literature, e.g., McIlhatton et al. (2016); de La Paz et al. (2022).

(iii) The negative impact of crime is greatest in the lowest priced residences, consistent with the results in McIlhatton et al. (2016).

With respect to our second hypothesis we show that certain demographic variables significantly influence housing prices. We estimate that population and % old (% over 65) levels, levels and changes in White and Asian population have a positive effect, while change in population and % old have a negative effect.

This study has attempted to shed some light on the very intricate issues relating various measures of crime to the prices of residential real estate. Our results should be interpreted in light of several important limitations. The entire analysis is based on data from Seattle over the time period 2008 to the end of 2020. The unique geographic characteristics of Seattle (in particular its relation to the shorelines and mountains), the idiosyncratic nature of its policing efforts (for example, the staffing reductions over the latter part of our sample) and its distinctive population dynamics (including the homelessness crisis) undoubtedly have a strong influence on our empirical work.

Table 8 Robustness check – quantile regression results using 911 calls with high priority

Panel A: Using *911 Calls (High Priority)*, without spatial autocorrelation adjustments						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Living area	0.318***	0.284***	0.306***	0.320***	0.340***	0.347***
	(0.004)	(0.006)	(0.003)	(0.003)	(0.004)	(0.006)
Lot area	0.021***	0.007***	0.013***	0.019***	0.026***	0.038***
	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Bedrooms	−0.019***	−0.008**	−0.012***	−0.012***	−0.014***	−0.019***
	(0.002)	(0.004)	(0.002)	(0.001)	(0.001)	(0.003)
Bathrooms (Full)	0.003	−0.010***	0.002	0.006***	0.004*	0.009***
	(0.002)	(0.004)	(0.003)	(0.002)	(0.002)	(0.003)
Age	−0.000***	−0.001***	−0.000***	−0.000***	−0.000**	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.095***	−0.064***	−0.077***	−0.082***	−0.088***	−0.100***
	(0.008)	(0.005)	(0.002)	(0.003)	(0.003)	(0.003)
Renovated	−0.046***	−0.269***	−0.107***	−0.007	0.026***	0.049***
	(0.011)	(0.017)	(0.012)	(0.006)	(0.004)	(0.010)
Fireplace	0.040***	0.059***	0.042***	0.029***	0.021***	0.013***
	(0.005)	(0.008)	(0.002)	(0.001)	(0.001)	(0.003)
Lake/Mountain view	0.104***	0.066***	0.069***	0.080***	0.103***	0.173***
	(0.005)	(0.007)	(0.004)	(0.003)	(0.006)	(0.007)
Population	0.017**	−0.019**	0.004	0.017***	0.025***	0.030***
	(0.007)	(0.009)	(0.005)	(0.004)	(0.004)	(0.007)
% Male population	0.907***	1.745***	1.120***	0.870***	0.647***	1.337***
	(0.229)	(0.183)	(0.102)	(0.088)	(0.091)	(0.209)
% White population	0.747***	0.978***	0.762***	0.756***	0.751***	0.728***
	(0.042)	(0.162)	(0.021)	(0.024)	(0.018)	(0.020)
% Asian population	0.008	0.007	−0.002	0.077**	0.146***	0.232***
	(0.053)	(0.130)	(0.022)	(0.032)	(0.023)	(0.040)
% Old population	0.268***	0.381***	0.091***	0.059**	0.065*	0.416***
	(0.038)	(0.083)	(0.032)	(0.027)	(0.035)	(0.054)
Number of crimes (IV)	−1.255**	−0.900	−1.298***	−1.062***	−0.994***	−1.371***
	(0.534)	(0.776)	(0.185)	(0.163)	(0.109)	(0.163)
Distance to crime hotspot	−0.073***	−0.073***	−0.068***	−0.070***	−0.074***	−0.085***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	12.241***	11.742***	12.183***	12.235***	12.341***	12.109***
	(0.102)	(0.136)	(0.053)	(0.035)	(0.050)	(0.071)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902
Panel B: Using *911 Calls (High Priority)*, with spatial autocorrelation adjustments						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.389***	0.402***	0.363***	0.360***	0.361***	0.373***
	(0.012)	(0.017)	(0.007)	(0.003)	(0.007)	(0.013)
Living area	0.261***	0.240***	0.252***	0.266***	0.285***	0.300***
	(0.004)	(0.006)	(0.003)	(0.003)	(0.002)	(0.005)

Table 8 (continued)

Lot area	0.018***	0.007***	0.013***	0.018***	0.022***	0.033***
	(0.001)	(0.002)	(0.000)	(0.001)	(0.001)	(0.001)
Bedrooms	−0.007***	−0.001	−0.002*	−0.002**	−0.002*	−0.009**
	(0.001)	(0.004)	(0.001)	(0.001)	(0.001)	(0.004)
Bathrooms (Full)	0.008***	−0.007***	0.006***	0.009***	0.009***	0.014***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Age	−0.001***	−0.002***	−0.001***	−0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.077***	−0.055***	−0.065***	−0.067***	−0.072***	−0.085***
	(0.006)	(0.007)	(0.002)	(0.002)	(0.002)	(0.003)
Renovated	−0.028***	−0.264***	−0.102***	0.021***	0.048***	0.069***
	(0.011)	(0.013)	(0.013)	(0.007)	(0.002)	(0.011)
Fireplace	0.025***	0.045***	0.025***	0.013***	0.010***	0.006**
	(0.004)	(0.007)	(0.002)	(0.002)	(0.002)	(0.002)
Lake/Mountain view	0.062***	0.035***	0.037***	0.046***	0.059***	0.106***
	(0.005)	(0.006)	(0.003)	(0.003)	(0.003)	(0.004)
Population	0.003	−0.011	−0.001	0.005	0.006*	0.014***
	(0.003)	(0.014)	(0.003)	(0.004)	(0.003)	(0.005)
% Male population	−0.048	0.097	0.171***	0.144**	0.115	0.496***
	(0.098)	(0.226)	(0.065)	(0.068)	(0.088)	(0.164)
% White population	0.575***	0.844***	0.606***	0.543***	0.502***	0.443***
	(0.020)	(0.072)	(0.018)	(0.014)	(0.018)	(0.018)
% Asian population	0.122***	0.217**	0.088***	0.120***	0.135***	0.138***
	(0.018)	(0.092)	(0.030)	(0.025)	(0.019)	(0.029)
% Old population	−0.173***	−0.187***	−0.188***	−0.213***	−0.228***	−0.203***
	(0.025)	(0.068)	(0.025)	(0.026)	(0.017)	(0.034)
Number of crimes (IV)	0.409**	0.943**	0.386***	0.188**	0.016	−0.285**
	(0.191)	(0.371)	(0.109)	(0.088)	(0.131)	(0.115)
Distance to crime hotspot	−0.039***	−0.037***	−0.038***	−0.040***	−0.042***	−0.047***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	7.530***	6.979***	7.728***	7.854***	7.934***	7.668***
	(0.176)	(0.202)	(0.085)	(0.078)	(0.073)	(0.175)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902

This table reports the results of 2SLS regression (model 1, as estimated in Eq. 1 and Eq. 3) and instrumental quantile regressions (models 2 – 6, as estimated in Eqs. 2 and 4) for 5, 25, 50, 75 and 95 percent quantiles, respectively. *Number of crimes* is instrumented by *Number of 911 calls (High Priority)*, defined as the number of 911 calls with priority levels of 1 (highest) or 2 (second highest), made in a given year per police beat. Panel A (B) performs 2SLS and quantile regressions without (with) spatial autocorrelation adjustments. Definitions of other variables are provided in Appendix 1 Table 10. Standard errors are reported in parentheses: for 2SLS regressions (model 1), standard errors are clustered by year and month. For quantile regressions (models 2 – 6), standard errors are obtained through bootstrap replications. *, ** and *** denote significance at the 10, 5 and 1 percent levels, respectively

Table 9 Robustness check – quantile regression results including distance to crime coldspot

Panel A: Using *Number of 911 Calls* as an IV, without spatial autocorrelation adjustments						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Living area	0.318***	0.285***	0.306***	0.321***	0.339***	0.348***
	(0.003)	(0.005)	(0.002)	(0.003)	(0.003)	(0.005)
Lot area	0.022***	0.009***	0.014***	0.020***	0.026***	0.038***
	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Bedrooms	−0.019***	−0.008*	−0.011***	−0.013***	−0.014***	−0.018***
	(0.002)	(0.005)	(0.002)	(0.002)	(0.002)	(0.003)
Bathrooms (Full)	0.004*	−0.009**	0.003	0.006***	0.005**	0.011***
	(0.002)	(0.004)	(0.002)	(0.002)	(0.002)	(0.004)
Age	−0.000***	−0.001***	−0.000***	−0.000***	−0.000**	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.092***	−0.063***	−0.073***	−0.079***	−0.087***	−0.099***
	(0.007)	(0.005)	(0.003)	(0.003)	(0.003)	(0.005)
Renovated	−0.046***	−0.278***	−0.110***	−0.003	0.023***	0.046***
	(0.011)	(0.023)	(0.014)	(0.008)	(0.007)	(0.013)
Fireplace	0.039***	0.062***	0.038***	0.027***	0.020***	0.013***
	(0.005)	(0.007)	(0.002)	(0.002)	(0.001)	(0.002)
Lake/Mountain view	0.109***	0.066***	0.074***	0.085***	0.106***	0.180***
	(0.006)	(0.007)	(0.004)	(0.004)	(0.006)	(0.009)
Population	0.014***	−0.019	0.001	0.016***	0.022***	0.026***
	(0.005)	(0.013)	(0.006)	(0.002)	(0.003)	(0.005)
% Male population	0.614***	1.484***	0.858***	0.629***	0.452***	1.008***
	(0.133)	(0.272)	(0.115)	(0.117)	(0.079)	(0.241)
% White population	0.776***	1.106***	0.801***	0.754***	0.758***	0.738***
	(0.037)	(0.154)	(0.023)	(0.017)	(0.021)	(0.023)
% Asian population	0.020	0.111	−0.000	0.067***	0.136***	0.221***
	(0.032)	(0.107)	(0.024)	(0.016)	(0.018)	(0.046)
% Old population	0.329***	0.427***	0.196***	0.119***	0.127***	0.376***
	(0.041)	(0.101)	(0.037)	(0.021)	(0.027)	(0.045)
Number of crimes (IV)	−0.694**	0.093	−0.588***	−0.713***	−0.659***	−1.007***
	(0.305)	(0.744)	(0.164)	(0.097)	(0.120)	(0.202)
Distance to crime hotspot	−0.072***	−0.071***	−0.067***	−0.069***	−0.073***	−0.084***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Distance to crime coldspot	−0.010***	−0.011***	−0.011***	−0.009***	−0.008***	−0.006***
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
Constant	12.359***	11.687***	12.260***	12.353***	12.443***	12.282***
	(0.067)	(0.196)	(0.090)	(0.057)	(0.046)	(0.078)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,898	61,898	61,898	61,898	61,898	61,898
Panel B: Using *Number of 911 Calls* as an IV, with spatial autocorrelation adjustments						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.389***	0.409***	0.364***	0.359***	0.360***	0.371***
	(0.013)	(0.014)	(0.009)	(0.008)	(0.007)	(0.011)

Table 9 (continued)

Living area	0.260***	0.239***	0.250***	0.267***	0.285***	0.299***
	(0.004)	(0.005)	(0.002)	(0.002)	(0.003)	(0.005)
Lot area	0.019***	0.008***	0.014***	0.018***	0.022***	0.034***
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.002)
Bedrooms	−0.007***	0.000	−0.001	−0.002	−0.003*	−0.008***
	(0.001)	(0.005)	(0.002)	(0.001)	(0.001)	(0.003)
Bathrooms (Full)	0.009***	−0.005	0.007***	0.010***	0.009***	0.016***
	(0.001)	(0.004)	(0.002)	(0.001)	(0.002)	(0.003)
Age	−0.001***	−0.002***	−0.001***	−0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.076***	−0.054***	−0.063***	−0.066***	−0.072***	−0.084***
	(0.005)	(0.004)	(0.002)	(0.003)	(0.003)	(0.004)
Renovated	−0.029***	−0.265***	−0.104***	0.020***	0.048***	0.070***
	(0.011)	(0.016)	(0.015)	(0.007)	(0.007)	(0.011)
Fireplace	0.025***	0.043***	0.024***	0.012***	0.010***	0.007***
	(0.004)	(0.006)	(0.003)	(0.002)	(0.002)	(0.002)
Lake/Mountain view	0.065***	0.040***	0.039***	0.048***	0.063***	0.110***
	(0.005)	(0.005)	(0.003)	(0.004)	(0.004)	(0.008)
Population	0.001	−0.015	−0.003	0.004	0.004	0.014***
	(0.002)	(0.014)	(0.005)	(0.003)	(0.004)	(0.005)
% Male population	−0.217**	−0.322	0.010	0.036	−0.003	0.332***
	(0.094)	(0.249)	(0.126)	(0.077)	(0.100)	(0.095)
% White population	0.592***	0.873***	0.618***	0.541***	0.507***	0.453***
	(0.030)	(0.067)	(0.020)	(0.012)	(0.017)	(0.019)
% Asian population	0.129***	0.210**	0.090***	0.107***	0.133***	0.140***
	(0.018)	(0.092)	(0.027)	(0.017)	(0.021)	(0.022)
% Old population	−0.138***	−0.224***	−0.157***	−0.175***	−0.205***	−0.211***
	(0.032)	(0.076)	(0.041)	(0.019)	(0.020)	(0.041)
Number of crimes (IV)	0.734***	1.338***	0.612***	0.354***	0.195**	−0.072
	(0.148)	(0.234)	(0.167)	(0.111)	(0.096)	(0.126)
Distance to crime hotspot	−0.039***	−0.036***	−0.038***	−0.040***	−0.042***	−0.046***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Distance to crime coldspot	−0.006***	−0.007***	−0.006***	−0.005***	−0.005***	−0.004***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)
Constant	7.602***	7.085***	7.801***	7.929***	8.008***	7.761***
	(0.174)	(0.251)	(0.115)	(0.102)	(0.129)	(0.176)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,898	61,898	61,898	61,898	61,898	61,898

This table reports the results of 2SLS regressions, as well as those of quantile regressions by adding *the Distance to crime coldspot* variable to the estimations. Panel A (B) performs 2SLS and quantile regressions without (with) spatial autocorrelation adjustments. *Distance to crime coldspot* is defined as the straight mile distance to the nearest crime coldspot, as identified by the Getis-Ord statistics. Definitions of other variables are provided in Appendix 1 Table 10. Standard errors are reported in parentheses: for 2SLS regressions (model 1), standard errors are clustered by year and month. For quantile regressions (models 2 – 6), standard errors are obtained through bootstrap replications. *, ** and *** denote significance at the 10, 5 and 1 percent levels, respectively

These limitations suggest several topics for further research, in particular the impact of policing efforts. As noted, enforcement levels are related to crime rates and to the frequency of 911 calls, complicating the already intricate relations between crime and the pricing of real estate. In addition, the link between actual crimes and 911calls requires additional analysis.

Appendix 1

Table 10 Variable definitions

Variable	Definition
Transaction prices	
Sale price	Sale price of property ($)
Log (Sale Price)	Log of sale price
WY	The weighted average sale price ($) of neighbor properties within 2000 feet
Log (WY)	Log of WY
Property-level characteristics	
Living area	Total living area in ft^2
Lot area	Total lot area in ft^2
Bedrooms	# of bedrooms
Bathrooms	# of full-size bathrooms
One story	A dummy variable equal to 1 for one-story property; zero otherwise
Year built	Year when property was built
Renovated	A dummy variable equal to 1 if previously renovated; zero otherwise
Age	Year of sale—Year built
Fireplace	A dummy variable equal to 1 for property with fireplace; zero otherwise
Environmental—natual views	
Lake/Mountain view	A dummy variable equal to 1 for property with natural view (Mt Rainier, Cascades, Puget Sound, Lake Washington, Olympics, Others); zero otherwise
Crime variables	
Number of crimes	Number of reported crimes per census tract, divided by population per census tract
Change in crime #	% change in number of reported crime from previous year (t-1) per census tract, or $\log\left(1 + number\ of\ crimes_t\right) - \log\left(1 + number\ of\ crimes_{t-1}\right)$

Table 10 (continued)

Variable	Definition
Distance to Crime Hotspot	
Straight miles (Nearest, 2nd, 3rd, 4th or 5th)	Straight mile distance to the nearest, 2nd, 3rd, 4th or 5th crime hotspot
Travel miles (Nearest, 2nd, 3rd, 4th or 5th)	Travel mile distance to the nearest, 2nd, 3rd, 4th or 5th crime hotspot
Travel minutes(Nearest, 2nd, 3rd, 4th or 5th)	Travel time (in minutes) to the nearest, 2nd, 3rd, 4th or 5th crime hotspot
911 calls	
Number of 911 calls	Log of (1 + number of 911 calls made per police beat) in year t
Change in 911 calls	% change in number of 911 calls from previous year (t-1) per police beat, or $\log\left(1 + \textit{Number of } 911 \textit{ Calls}_t\right) - \log(1 + \textit{Number of } 911 \textit{ Calls}_{t-1})$
Sociodemographic variables	
Population	Log of population per census tract
Change in population	% change in population from previous year per census tract, or $\log\left(\textit{Population}_t\right) - \log\left(\textit{Population}_{t-1}\right)$
% Male population	Percent of male population per census tract
Change in male population	% change in male population from previous year per census tract, or $\log\left(\textit{Male Population}_t\right) - \log\left(\textit{Male Population}_{t-1}\right)$
% Female population	Percent of female population per census tract
Change in female population	% change in female population from previous year per census tract, or $\log\left(\textit{Female Population}_t\right) - \log(\textit{Female Population}_{t-1})$
% White population	Percent of White population per census tract
Change in white population	% change in White population from previous year per census tract, or $\log\left(\textit{White Population}_t\right) - \log\left(\textit{White Population}_{t-1}\right)$
% Asian population	Percent of Asian population per census tract
Change in Asian population	% change in Asian population from previous year per census tract, or $\log\left(\textit{Asian Population}_t\right) - \log(\textit{Asian Population}_{t-1})$
% Black population	Percent of black population per census tract
Change in black population	% change in black population from previous year per census tract, or $\log\left(\textit{Black Population}_t\right) - \log(\textit{Black Population}_{t-1})$
% Old population	Percent of population over age 65 per census tract
Change in old population	% change in old population from previous year per census tract, or $\log\left(\textit{Old Population}_t\right) - \log(\textit{Old Population}_{t-1})$

Appendix 2

Table 11 Baseline results with zip code fixed effects

Panel A: Without spatial autocorrelation adjustments						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Living area	0.288***	0.255***	0.280***	0.294***	0.306***	0.312***
	(0.004)	(0.006)	(0.003)	(0.002)	(0.003)	(0.004)
Lot area	0.026***	0.016***	0.021***	0.024***	0.030***	0.041***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Bedrooms	−0.016***	−0.003	−0.009***	−0.010***	−0.011***	−0.017***
	(0.002)	(0.003)	(0.002)	(0.002)	(0.001)	(0.003)
Bathrooms (Full)	0.007***	0.001	0.006***	0.008***	0.008***	0.013***
	(0.002)	(0.004)	(0.002)	(0.002)	(0.003)	(0.004)
Age	−0.001***	−0.001***	−0.001***	−0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.076***	−0.053***	−0.058***	−0.065***	−0.075***	−0.086***
	(0.006)	(0.006)	(0.003)	(0.003)	(0.004)	(0.004)
Renovated	−0.033***	−0.265***	−0.106***	0.018**	0.041***	0.052***
	(0.011)	(0.015)	(0.010)	(0.009)	(0.004)	(0.009)
Fireplace	0.020***	0.036***	0.021***	0.014***	0.006***	−0.006**
	(0.005)	(0.005)	(0.002)	(0.002)	(0.001)	(0.003)
Lake/Mountain view	0.140***	0.094***	0.107***	0.114***	0.138***	0.211***
	(0.006)	(0.004)	(0.003)	(0.003)	(0.006)	(0.010)
Population	0.073***	0.080***	0.075***	0.071***	0.073***	0.087***
	(0.008)	(0.007)	(0.005)	(0.003)	(0.006)	(0.014)
% Male population	0.355*	0.304*	0.233	0.296**	0.627***	1.935***
	(0.188)	(0.162)	(0.145)	(0.127)	(0.114)	(0.283)
% White population	0.261***	0.262***	0.285***	0.250***	0.242***	0.104
	(0.043)	(0.083)	(0.032)	(0.032)	(0.036)	(0.065)
% Asian population	−0.493***	−0.437***	−0.399***	−0.418***	−0.405***	−0.576***
	(0.060)	(0.078)	(0.052)	(0.033)	(0.028)	(0.061)
% Old population	0.664***	0.246***	0.283***	0.493***	0.678***	1.148***
	(0.060)	(0.054)	(0.029)	(0.048)	(0.031)	(0.183)
Number of crimes (IV)	−1.849***	−3.070***	−2.646***	−2.263***	−1.898***	−1.870***
	(0.376)	(0.515)	(0.185)	(0.180)	(0.209)	(0.336)
Distance to crime hotspot	−0.071***	−0.082***	−0.074***	−0.071***	−0.068***	−0.065***
	(0.004)	(0.005)	(0.001)	(0.001)	(0.002)	(0.002)
Constant	12.432***	12.398***	12.487***	12.507***	12.330***	11.711***
	(0.113)	(0.097)	(0.074)	(0.058)	(0.059)	(0.212)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
Zip area dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902
Panel B: With spatial autocorrelation adjustments						
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Log (Sale Price)	2SLS	Q: 0.05	Q: 0.25	Q: 0.50	Q: 0.75	Q: 0.95
Log (WY)	0.292***	0.282***	0.271***	0.265***	0.262***	0.266***
	(0.013)	(0.018)	(0.009)	(0.006)	(0.010)	(0.017)

Table 11 (continued)

Living area	0.256***	0.231***	0.249***	0.265***	0.281***	0.290***
	(0.004)	(0.005)	(0.002)	(0.002)	(0.002)	(0.004)
Lot area	0.023***	0.012***	0.018***	0.022***	0.027***	0.038***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)
Bedrooms	−0.008***	−0.001	−0.002**	−0.003*	−0.004***	−0.008***
	(0.001)	(0.004)	(0.001)	(0.002)	(0.001)	(0.003)
Bathrooms (Full)	0.009***	0.000	0.008***	0.010***	0.009***	0.015***
	(0.001)	(0.004)	(0.002)	(0.002)	(0.002)	(0.002)
Age	−0.001***	−0.002***	−0.001***	−0.001***	−0.001***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
One story	−0.070***	−0.054***	−0.058***	−0.060***	−0.067***	−0.080***
	(0.005)	(0.005)	(0.003)	(0.003)	(0.002)	(0.004)
Renovated	−0.025**	−0.258***	−0.102***	0.028***	0.053***	0.067***
	(0.011)	(0.013)	(0.014)	(0.008)	(0.003)	(0.007)
Fireplace	0.016***	0.032***	0.015***	0.008***	0.002	−0.001
	(0.004)	(0.005)	(0.002)	(0.002)	(0.002)	(0.004)
Lake/Mountain view	0.095***	0.059***	0.066***	0.076***	0.094***	0.158***
	(0.005)	(0.005)	(0.003)	(0.002)	(0.004)	(0.006)
Population	0.031***	0.056***	0.045***	0.036***	0.028***	0.029***
	(0.004)	(0.017)	(0.003)	(0.006)	(0.007)	(0.010)
% Male population	−0.315***	0.089	−0.090	−0.137	−0.061	0.210
	(0.104)	(0.100)	(0.111)	(0.089)	(0.117)	(0.273)
% White population	0.359***	0.270***	0.309***	0.309***	0.334***	0.299***
	(0.023)	(0.076)	(0.026)	(0.036)	(0.021)	(0.045)
% Asian population	−0.149***	−0.200***	−0.159***	−0.151***	−0.122***	−0.166***
	(0.034)	(0.060)	(0.033)	(0.027)	(0.032)	(0.046)
% Old population	0.098**	0.009	0.023	0.088***	0.143***	0.198**
	(0.039)	(0.054)	(0.031)	(0.026)	(0.034)	(0.094)
Number of crimes (IV)	−0.055	−1.430***	−0.956***	−0.632***	−0.249**	−0.124
	(0.192)	(0.521)	(0.196)	(0.195)	(0.099)	(0.274)
Distance to crime hotspot	−0.045***	−0.056***	−0.051***	−0.048***	−0.046***	−0.044***
	(0.004)	(0.004)	(0.002)	(0.001)	(0.001)	(0.002)
Constant	8.929***	8.748***	9.057***	9.237***	9.280***	9.186***
	(0.197)	(0.267)	(0.147)	(0.083)	(0.107)	(0.120)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
Zip area dummies	Yes	Yes	Yes	Yes	Yes	Yes
N	61,902	61,902	61,902	61,902	61,902	61,902

This table reports the results of baseline regressions (as estimated in Eq. 1 – 4) with zip code fixed effects. Panel A reports the results from Eq. 1 (2SLS) and 2 (quantile regressions) without spatial autocorrelation adjustments. Panel B reports the results from Eq. 3 (2SLS) and 4 (instrumental variable quantile regressions) with spatial autocorrelation adjustments. Standard errors are reported in parentheses: for 2SLS regressions (model 1), standard errors are clustered by year and month. For quantile regressions (models 2 – 6), standard errors are obtained through bootstrap replications. *, ** and *** denote significance at the 10, 5 and 1 percent levels, respectively

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Data Availability The data analyzed in this study come from the King County Assessor's website (https://kingcounty.gov/depts/assessor.aspx), the Washington State Office of Financial Management website (https://ofm.wa.gov/), the Seattle Police Department website (https://www.seattle.gov/police), and the Seattle Open Data Portal (https://data.seattle.gov/). The geographic data were analyzed using Maptitude Mapping Software (Caliper Corporation), GeoDa (https://geodacenter.github.io/), and Texas A&M University Geoservices (https://geoservices.tamu.edu/). The data from Caliper Corporation were used under license.

Code Availability Not applicable.

Declarations

Conflicts of Interests/Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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References

Abbott, J. K., & Klaiber, H. A. (2011). An embarrassment of riches: Confronting omitted variable bias and multi-scale capitalization in hedonic price models. *Review of Economics and Statistics, 93*, 1331–1342. https://doi.org/10.1162/REST_a_00134

Andresen, M. A., & Malleson, N. (2013). Crime seasonality and its variations across space. *Applied Geography, 43*, 25–35. https://doi.org/10.1016/j.apgeog.2013.06.007

Anselin, L., Cohen, J., Cook, D., Gorr, W., & Tita, G. (2000). Spatial analyses of crime. *Criminal Justice, 4*, 213–262.

Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy, 76*, 169–217.

Boggess, L. N., Greenbaum, R. T., & Tita, G. E. (2013). Does crime drive housing sales? Evidence from Los Angeles. *Journal of Crime and Justice, 36*, 299–318. https://doi.org/10.1080/0735648X.2013.812976

Braakmann, N. (2012). The link between non-property crime and house prices: Evidence from UK street-level data. MPRA Paper No. 44884.

Braga, A. A., Papachristos, A. V., & Hureau, D. M. (2014). The effects of hotspots policing on crime: An updated systematic review and meta-analysis. *Justice Quarterly, 31*, 633–663. https://doi.org/10.1080/07418825.2012.673632

Brantingham, P. L., & Brantingham, P. J. (1982). Mobility, notoriety and crime: A study of crime patterns in urban nodal points. *Journal of Environmental Systems, 11*, 89–99.

Brooks, T. J., Humphreys, B. R., & Nowak, A. (2020). Strip clubs, "secondary effects" and residential property prices. *Real Estate Economics, 48*, 850–885. https://doi.org/10.1111/1540-6229.12236

Burgess, J. F., & Harmon, O. R. (1991). Specification tests in hedonic models. *The Journal of Real Estate Finance and Economics, 4*, 375–393. https://doi.org/10.1007/BF00219505

Case, K. E., & Mayer, C. J. (1996). Housing price dynamics within a metropolitan area. *Regional Science and Urban Economics, 26*, 387–407. https://doi.org/10.1016/0166-0462(95)02121-3

Ceccato, V. (2015). Vandalism. In W. G. Jennings (Ed.), *The encyclopedia of crime and punishment.* Wiley Online Library.

Ceccato, V., & Wilhelmsson, M. (2011). The impact of crime on apartment prices: Evidence from Stockholm, Sweden. *Geografiska Annaler: Series B, Human Geography, 93*, 81–103. https://doi.org/10.1111/j.1468-0467.2011.00362.x

Ceccato, V., & Wilhelmsson, M. (2018). Does crime impact real estate prices? An assessment of accessibility and location. In J. Gerben, N. Bruinsma, & S. D. Johnson (Eds.), *Oxford handbook on environmental criminology* (pp. 518–544). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780190279707.013.30

Ceccato, V., & Wilhelmsson, M. (2020). Do crime hotspots affect housing prices? *Nordic Journal of Criminology, 21*, 84–102. https://doi.org/10.1080/2578983X.2019.1662595

Chamberlain, A. W., & Boggess, L. N. (2016). Relative difference and burglary location: Can ecological characteristics of a burglar's home neighborhood predict offense location? *Journal of Research in Crime and Delinquency, 53*, 872–906. https://doi.org/10.1177/0022427816647993

Cohen, M. A. (1990). A note on the cost of crime to victims. *Urban Studies, 27*, 139–146. https://doi.org/10.1080/00420989020080081

Cullen, J. B., & Levitt, S. D. (1999). Crime, urban flight, and the consequences for cities. *Review of Economics and Statistics, 99*, 159–169. https://doi.org/10.1162/003465399558030

Curry, G. D., & Spergel, I. A. (1988). Gang homicide, delinquency, and community. *Criminology, 26*, 381–406. https://doi.org/10.1111/j.1745-9125.1988.tb00847.x

de La Paz, P. T., Berry, J., McIlhatton, D., Chapman, D., & Bergonzoli, K. (2022). The impact of crimes on house prices in LA County. *Journal of European Real Estate Research, 15*, 88–111. https://doi.org/10.1108/JERER-01-2021-0002

Deng, Y., McMillen, D. P., & Sing, T. F. (2014). Matching indices for thinly-traded commercial real estate in Singapore. *Regional Science and Urban Economics, 47*, 86–98. https://doi.org/10.1016/j.regsciurbeco.2013.09.017

Gehlke, C. E., & Biehl, K. (1934). Certain effects of grouping upon the size of the cor- relation coefficient in census tract material. *Journal of the American Statistical Association., 29*, 169–170. https://doi.org/10.1080/01621459.1934.10506247

Getis, A. (2010). Spatial autocorrelation. In *Handbook of applied spatial analysis* (pp. 255–278). Springer. https://doi.org/10.1007/978-3-642-03647-7_14

Getis, A., & Ord, J. (1992). The analysis of spatial association by use of distance statistics. *Geographical Analysis, 24*, 189–206. https://doi.org/10.1111/j.1538-4632.1992.tb00261.x

Gibbons, S. (2004). The costs of urban property crime. *The Economic Journal, 114*, F441–F463. https://doi.org/10.1111/j.1468-0297.2004.00254.x

Glaeser, E. L., & Sacerdote, B. (1999). Why is there more crime in cities? *Journal of Political Economy, 107*, S225–S258. https://doi.org/10.1086/250109

Goodman, A. C. (1998). Andrew Court and the invention of hedonic price analysis. *Journal of Urban Economics, 44*, 291–298. https://doi.org/10.1006/juec.1997.2071

Goudriaan, H., Wittebrood, K., & Nieuwbeerta, P. (2006). Neighbourhood characteristics and reporting crime: Effects of social cohesion, confidence in police effectiveness and socio- economic disadvantage. *British Journal of Criminology, 46*, 719–742. https://doi.org/10.1093/bjc/azi096

Guerrieri, V., Hartley, D., & Hurst, E. (2012). Within-city variation in urban decline: The case of Detroit. *American Economic Review, 102*, 120–126. https://doi.org/10.1257/aer.102.3.120

Guerrieri, V., Hartley, D., & Hurst, E. (2013). Endogenous gentrification and housing price dynamics. *Journal of Public Economics, 100*, 45–60. https://doi.org/10.1016/j.jpubeco.2013.02.001

Guerry, A., Lacroix, S. F., de Silvestre, A.-F., & Girard, P.-S. (1833). *Essai sur la statistiquemorale de la France*. University of Gent.

Hale, C. (1996). Fear of crime: A review of the literature. *International Review of Victimology, 4*, 79–150. https://doi.org/10.1177/026975809600400201

Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica, 46*, 1251–1271. https://doi.org/10.2307/1913827

Hellman, D. A., & Naroff, J. L. (1979). The impact of crime on urban residential property values. *Urban Studies, 16*, 105–112. https://doi.org/10.1080/713702454

Herath, S. K., & Maier, G. (2010). *The hedonic price method in real estate and housing market research. A review of the literature. Institute for Regional Development and Environment* (pp. 1–21). University of Economics and Business.

Hipp, J. R. (2011a). Spreading the wealth: The effect of the distribution of income and race/ethnicity across households and neighborhoods on city crime trajectories. *Criminology, 49*, 631–665. https://doi.org/10.1111/j.1745-9125.2011.00238.x

Hipp, J. R. (2011b). Violent crime, mobility decisions, and neighborhood racial/ethnic transition. *Social Problems, 58*, 410–432. https://doi.org/10.1525/sp.2011.58.3.410

Hui, E. C., Chau, C. K., Pun, L., & Law, M. Y. (2007). Measuring the neighboring and environmental effects on residential property value: Using spatial weighting matrix. *Building and Environment, 42*, 2333–2343. https://doi.org/10.1016/j.buildenv.2006.05.004

Hwang, J. (2020). Gentrification without segregation? Race, immigration, and renewal in a diversifying city. *City & Community, 19*(3), 538–572. https://doi.org/10.1111/cico.12419

Ihlanfeldt, K., & Mayock, T. (2010a). Panel data estimates of the effects of different types of crime on housing prices. *Regional Science and Urban Economics, 40*, 161–172. https://doi.org/10.1016/j.regsciurbeco.2010.02.005

Ihlanfeldt, K., & Mayock, T. (2010b). Property values and crime. In B. Benson & P. Zimmerman (Eds.), *Handbook on the economics of crime.* Edward Elgar.

Kain, J. F., & Quigley, J. M. (1970). Measuring the value of housing quality. *Journal of the American Statistical Association, 65*, 532–548.

Kelly, M. (2000). Inequality and crime. *Review of Economics and Statistics, 82*, 530–539. https://doi.org/10.1162/003465300559028

Kim, C. W., Phipps, T. T., & Anselin, L. (2003). Measuring the benefits of air quality improvement: A spatial hedonic approach. *Journal of Environmental Economics and Management, 45*(1), 24–39. https://doi.org/10.1016/S0095-0696(02)00013-X

Koenker, R., & Hallock, K. F. (2001). Quantile regression. *Journal of Economic Perspectives, 15*, 143–156. https://doi.org/10.1257/jep.15.4.143

Kreager, D. A., Lyons, C. J., & Hays, Z. R. (2011). Urban revitalization and Seattle crime, 1982–2000. *Social Problems, 58*, 615–639. https://doi.org/10.1525/sp.2011.58.4.615

Larsen, J. E., Lowrey, K. J., & Coleman, J. W. (2003). The effect of proximity to a registered sex offender's residence on single-family house selling price. *The Appraisal Journal, 71*, 253.

Linden, L., & Rockoff, J. E. (2008). Estimates of the impact of crime risk on property values from Megan's laws. *American Economic Review, 98*, 1103–1127. https://doi.org/10.1257/aer.98.3.1103

Lynch, A. K., & Rasmussen, D. W. (2001). Measuring the impact of crime on house prices. *Applied Economics, 33*, 1981–1989. https://doi.org/10.1080/00036840110021735

MacDonald, Z. (2001). Revisiting the dark figure: A microeconometric analysis of the under-reporting of property crime and its implications. *British Journal of Criminology, 41*, 127–149. https://doi.org/10.1093/bjc/41.1.127

MacDonald, Z. (2002). Official crime statistics: Their use and interpretation. *The Economic Journal, 112*, F85–F106. https://doi.org/10.1111/1468-0297.00685

Malpezzi, S. (2003). Hedonic pricing models: A selective and applied review. In T. O'Sullivan & K. Gibb (Eds.), *Housing economics and public policy* (pp. 67-89). Blackwell Science.

Matthews, S., Yang, T. C., Hayslett, K. L., & Ruback, R. B. (2010). Built environment and property crime in Seattle, 1998–2000: A Bayesian analysis. *Environment and Planning A, 42*, 1403–1420. https://doi.org/10.1068/a42393

McIlhatton, D., McGreal, W., de la Paz, P. T., & Adair, A. (2016). Impact of crime on spatial analysis of house prices: Evidence from a UK city. *International Journal of Housing Markets and Analysis, 9*, 627–647. https://doi.org/10.1108/IJHMA-10-2015-0065

Myers, S. L. (1980). Why are crimes underreported? What is the crime rate? Does it "really" matter? *Social Science Quarterly, 61*, 23–43.

Olea, J., & Pflueger, C. (2013). A robust test for weak instruments. *Journal of Business and Economic Statistics, 31*, 358–369. https://doi.org/10.1080/00401706.2013.806694
Pace, R. K., & LeSage, J. P. (2009). *Introduction to spatial econometrics*. CRC Press/Chapman & Hall/Taylor & Francis.
Pope, J. C. (2008). Fear of crime and housing prices: Household reactions to sex offender registries. *Journal of Urban Economics, 64*, 601–614. https://doi.org/10.1016/j.jue.2008.07.001
Pyle, D. (1998). Crime and unemployment: What do empirical studies show? *International Journal of Risk Security and Crime Prevention, 3*, 169–180.
Quetelet, A., 1833. *Recherches sur le Penchant au Crime aux Differens Ages*. Bruxelles, M. Hayez.
Rizzo, M. J. (1979). The cost of crime to victims: An empirical analysis. *The Journal of Legal Studies, 8*, 177–205.
Roncek, D. W. (1981). Dangerous places: Crime and residential environment. *Social Forces, 60*, 74–96. https://doi.org/10.1093/sf/60.1.74
Roncek, D. W., & Bell, R. (1981). Bars, blocks, and crimes. *Journal of Environmental Systems, 11*, 35–47. https://doi.org/10.2190/R0G0-FRWY-100J-6KTB
Roncek, D. W., & Maier, P. A. (1991). Bars, blocks, and crimes revisited: Linking the theory of routine activities to the empiricism of "hotspots." *Criminology, 29*, 725–753. https://doi.org/10.1111/j.1745-9125.1991.tb01086.x
Sampson, R. J., Raudenbush, S. W., & Earls, F. (1997). Neighborhoods and violent crime: A multilevel study of collective efficacy. *Science, 277*, 918–924. https://doi.org/10.1126/science.277.5328.918
Sampson, R. J., & Wilson, W. J. (2020). *Toward a theory of race, crime, and urban inequality*. Routledge.
Schwartz, A. E., Susin, S., & Voicu, I. (2003). Has falling crime driven New York City's real estate boom? *Journal of Housing Research*, *14*, 101–135. https://www.jstor.org/stable/44944776
Seattle Police Department. (2023). Crime Report 2022. Retrieved November 4, 2023, from https://www.seattle.gov/police/information-and-data/data/crime-dashboard
Shaw, C. R., & McKay, H. D. (1942). *Juvenile delinquency and urban areas*. University of Chicago Press.
Sherman, L. W., Gartin, P. R., & Buerger, M. E. (1989). Hotspots of predatory crime: Routine activities and the criminology of place. *Criminology, 27*, 27–56. https://doi.org/10.1111/j.1745-9125.1989.tb00862.x
Sirmans, S., Macpherson, D., & Zietz, E. (2005). The composition of hedonic pricing models. *Journal of Real Estate Literature, 13*, 1–44. https://doi.org/10.1080/10835547.2005.12090154
Skogan, W. G. (1984). Reporting crimes to the police: The status of world research. *Journal of Research in Crime and Delinquency, 21*, 113–137. https://doi.org/10.1177/0022427884021002003
Skogan, W. G. (1989). Communities, crime, and neighborhood organization. *Crime & Delinquency, 35*, 437–457. https://doi.org/10.1177/0011128789035003008
Spelman, W. (2013). Prisons and crime, backwards in high heels. *Journal of Quantitative Criminology, 29*, 643–674. https://doi.org/10.1007/s10940-013-9193-2
Tankebe, J. (2013). Viewing things differently: The dimensions of public perceptions of police legitimacy. *Criminology, 51*, 103–135. https://doi.org/10.1111/j.1745-9125.2012.00291.x
Thaler, R. (1978). A note on the value of crime control: Evidence from the property market. *Journal of Urban Economics, 5*, 137–145. https://doi.org/10.1016/0094-1190(78)90042-6
Velez, M. B., Lyons, C. J., & Boursaw, B. (2012). Neighborhood housing investments and violent crime in Seattle, 1981–2007. *Criminology, 50*, 1025–1056. https://doi.org/10.1111/j.1745-9125.2012.00287.x
Waltl, S. R. (2019). Variation across price segments and locations: A comprehensive quantile regression analysis of the Sydney housing market. *Real Estate Economics, 47*, 723–756. https://doi.org/10.1111/1540-6229.12177
Weisburd, D. (2015). The law of crime concentration and the criminology of place. *Criminology, 53*, 133–157. https://doi.org/10.1111/1745-9125.12070
Weisburd, D., Bushway, S., Lum, C., & Yang, S. M. (2004). Trajectories of crime at places: A longitudinal study of street segments in the city of Seattle. *Criminology, 42*, 283–322. https://doi.org/10.1111/j.1745-9125.2004.tb00521.x
Wickramasekera, N., Wright, J., Elsey, H., Murray, J., & Tubeuf, S. (2015). Cost of crime: A systematic review. *Journal of Criminal Justice, 43*, 218–228. https://doi.org/10.1016/j.jcrimjus.2015.04.009
Wilhelmsson, M., & Ceccato, V. (2015). Does burglary affect property prices in a nonmetropolitan municipality? *Journal of Rural Studies, 39*, 210–218. https://doi.org/10.1016/j.jrurstud.2015.03.014
Wilson, J. Q., & Kelling, G. L. (1982). Broken windows. *Atlantic Monthly, 249*, 29–38.
Wu, D. M. (1973). Alternative tests of independence between stochastic regressors and disturbances. *Econometrica, 41*, 733–750. https://doi.org/10.2307/1914093

Zhang, L. (2016). Flood hazards impact on neighborhood house prices: A spatial quantile regression analysis. *Regional Science and Urban Economics, 60*, 12–19. https://doi.org/10.1016/j.regsciurbeco.2016.06.005

Zietz, J., Zietz, E. N., & Sirmans, G. S. (2008). Determinants of house prices: A quantile regression approach. *The Journal of Real Estate Finance and Economics, 37*, 317–333. https://doi.org/10.1007/s11146-007-9053-7

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